Chapter 3

RESEARCH METHODOLOGY

This chapter discusses the methodology used in developing the Integrated Personnel Profiling and Monitoring System for the Dapitan City Police Station. It explains the research process, tools, and techniques applied to ensure that the system was designed, developed, and implemented effectively according to the needs of the organization. The researchers used the Rapid Application Development (RAD) methodology to guide the entire process from planning to implementation ensuring that the system was completed efficiently through continuous user feedback and prototype testing.

System Development Methodology

Rapid Application Development (RAD) Methodology

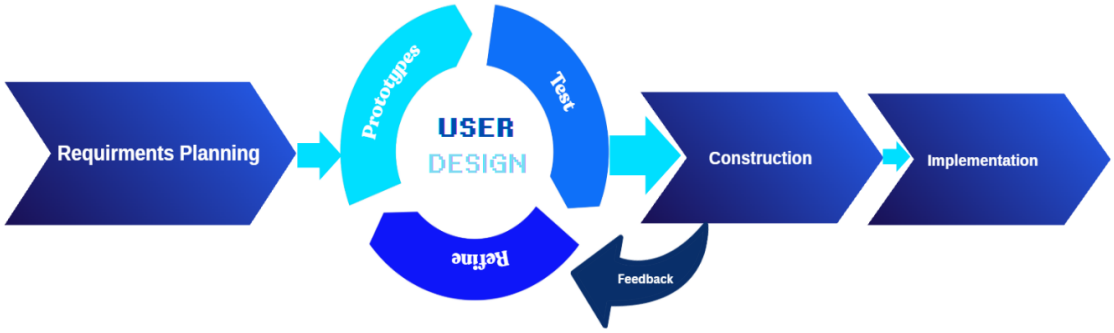


Figure 1. Rapid Application Development (RAD).

In this study, the methodology used was specifically designed by the researcher but still grounded in the principles of the Rapid Application Development (RAD) Model. The development focused on rapid system creation through continuous user feedback and fast prototyping. Rather than spending extensive time on detailed initial planning, prototypes were immediately built, presented to users, and refined based on their suggestions. This

approach allowed the development to address actual issues encountered in managing personnel data, seminars, and trainings. The customized model consists of four main phases: Planning, Design, Construction, and Implementation each guiding the smooth and intended development of the system.

Requirements Planning

During this phase, all necessary information was gathered directly from the Dapitan City Police Station. Problems with manual record-keeping were identified, such as difficulties in retrieving data, inaccurate or incomplete records, and the lack of proper monitoring of seminars attended by personnel. The main objectives of the system were also defined, including the centralization of all personnel data, automation of record management, and the capability to generate reports whenever needed. This phase established the overall direction and purpose of the system, ensuring that all planned components aligned with the actual needs of the police station.

User Design

The user design of the Integrated Personnel Profiling and Monitoring (IPPM) System was developed using the Rapid Application Development (RAD) approach to ensure that the interface aligned with the actual workflow of the Dapitan City Police Station. After determining the requirements and analyzing the system through diagrams such as the Context Diagram, DFD, Object Model, and ERD, interface prototypes representing key system windows: Login Window, Dashboard, Personnel List, Personnel Forms and Restore modules, Training and Award management, Report Filters, PDS Generator with watermark, and User Account functions were created.

These prototypes were presented to the users for review and hands-on testing. Their feedback guided improvements in layout clarity, button labels, navigation flow, and form organization. Multiple cycles of testing and refinement ensured that each feature such as saving personnel data, managing trainings and awards, uploading documents, generating reports, and printing the PDS functioned correctly and supported real operational needs. Improvements made during the design stage included enhanced search and filtering options, a purpose prompt and watermark for PDS printing, and a more secure Forgot Password process with OTP verification. Each update was validated by users to ensure usability and reliability.

Through continuous evaluation and refinement, the final user design became intuitive, efficient, and suitable for deployment. It reflects the actual needs of station users and provides a user-friendly interface that supports fast, accurate, and organized personnel management.

Prototypes

During the Design Phase of the Integrated Personnel Profiling and Monitoring System (IPPM), prototypes were developed to visualize the system's interface and functionality. These prototypes, which included key system forms and windows, were presented to the Dapitan City Police Station for review. Feedback was actively gathered from users to ensure that the design aligned with their workflow and operational needs. The insights collected were used to refine the prototypes, improving usability and addressing identified shortcomings. Testing was conducted afterward to confirm that all implemented changes met the defined requirements and functioned as intended, ensuring that the IPPM system was developed with the user's needs in mind.

Construction

The Construction Phase focused on the actual development of the system using tools such as PHP, HTML, CSS, JavaScript, and MySQL. Key features implemented included user authentication, personnel record management, seminar and training monitoring, and report generation. Throughout development, each function was tested to ensure accuracy and reliability. Any identified errors were immediately corrected to maintain system functionality. This phase ensured that the system would operate smoothly once fully implemented.

Implementation

The system was planned for implementation and underwent development and testing. After the development process was completed and the system became fully functional, it was installed at the Dapitan City Police Station for use in an offline environment. The implementation process was carried out smoothly with minimal disruptions to daily operations. Before implementation, users underwent proper training to ensure effective operation and management of the system. Additional testing and user evaluations were conducted to verify that all features functioned correctly and efficiently, even without an internet connection. After implementation, regular maintenance and updates were performed to maintain performance and support continuous improvement.

Requirement Specifications

This section outlines all the requirements needed for the Integrated Personnel Profiling and Monitoring System for the Dapitan City Police Station. These requirements serve as the foundation for the system’s design and validation. It also includes other important details needed to ensure that the system is properly designed and works as

expected. The requirements are divided into several categories: functional requirements, input requirements, output requirements, software requirements, and hardware requirements. Each category plays an important role in making sure the system is complete, efficient, and meets the needs of its users.

Functionality Requirements

This section presents the requirements needed to design, develop, and operate the system effectively. These include functional requirements, input and output requirements, and the hardware and software specifications necessary for operation.

Table 1. Functional Requirements

This section described the main operations such as managing personnel, user login, trainings, awards, reports, PDS generation, and activity logs. These functions ensure accurate record-keeping, efficient administration, and secure system operation.

Functional Requirements	Description
Manage User Login	Provides secure login authentication for the user to access and operate the system.
Manage Personnel	Allows the user to add, update or view personnel profiles. This includes storing personal information, rank, current designation, and service details of each police personnel.
Manage Training	Records and monitors seminars and training attended by personnel. It helps track participation, schedules, and certificates for organizations needed.

Manage Awards	Stores and tracks awards, recognitions, and commendations received by personnel. It provides an organized record for evaluation and reporting purposes.
Generate Reports	Automatically produces various reports such as personnel lists, training records, and award summaries. Reports can be exported or printed for administrative use.
Generate PDS (Personnel Data Sheet)	Automatically compiles and generates a complete personnel data sheet containing personal, professional, and service information, ready for printing or digital submission.
Manage Activity Logs	Records system actions and operations performed by the user for monitoring purposes.

Table 2. Input Requirements

Input requirements are the essential data and information that the Integrated Personnel Profiling and Monitoring System needs in order to function properly. These inputs serve as the foundation for the system's operations, enabling it to process, store, and manage various personnel-related tasks accurately and efficiently.

By ensuring that all required inputs are complete and accurate, the system can deliver reliable outputs and support effective decision-making.

Input Requirements	Description
Manage User Login	User enters login credentials (username and password) to access the system.
Manage Personnel	User inputs personnel information such as name, rank, date of birth, address, contact details, designation, and service record.
Manage Training	User encodes seminar or training details including course title, location, schedule, instructor, and participants.
Manage Awards	User inputs award details such as award name, date, issuing authority, and remarks.
Generates Reports	User selects report type, date range, or category (e.g., personnel, awards, training).
Generates PDS (Personal Data Sheet)	User selects a specific personnel record. System retrieves all related data (personal, service, awards, and training details).
Manage Activity Logs	System automatically records input details such as action performed, timestamp, and personnel involved whenever the user performs operations.

Table 3. Output Requirements

These are the main results or information that the system produces after processing the input data:

Output Requirements	Description
Manage User Login	Displays authentication results such as successful login, incorrect password notification, or access denied messages.
Manage Personnel	Displays updated personnel profiles, generates confirmation messages for successful add and update actions, and shows a searchable list of personnel records,

	ensuring that users can efficiently track changes and easily locate specific information whenever needed.
Manage Training	Displays a list of training attended by personnel. Generates printable or exportable training summaries and participation certificates.
Manage Awards	Displays award records linked to the corresponding personnel. Generates printable or exportable award summaries and recognition lists.
Generate Reports	System produces formatted reports that can be viewed or printed.
Generate PDS (Personal Data Sheet)	Generates a complete, printable Personnel Data Sheet (PDS) in standardized format ready for submission or printing.
Manage Activity Logs	Displays recorded system logs including action performed, exact time, and status for monitoring system usage and security tracking.

Table 4. Software Requirements

The following software components are essential for both the development and deployment teams, as well as for the user of the Integrated Personnel Profiling and Monitoring System for the Dapitan City Police Station. These tools ensure that the system operates smoothly, meets all technical requirements, and delivers the necessary functionality to support efficient and reliable system performance. By utilizing these software components, the system can maintain consistency, security, and optimal usability across all stages of its operation.

Development	Description
XAMPP	for local hosting and Database management

Visual Studio Code	for Coding and Development
HTML, CSS, JavaScript, Bootstrap	for frontend designs
PHP	for backend development
MySQL	for database management
Google Chrome or Microsoft Edge/	for accessing and testing the system interface

Table 5. Hardware Requirements

The hardware requirements table presents both the minimum and recommended specifications for the Integrated Personnel Profiling and Monitoring System of the Dapitan City Police Station. Although the system can function with the minimum specifications, utilizing the recommended hardware ensures better performance, faster processing, greater reliability, and an improved user experience, allowing the system to run more smoothly and efficiently.

Component	Minimum Specification	Recommended Specification	Description
Storage Drive	HDD (Hard Disk Drive) 500 GB	SDD (Solid State Drive) 500 GB or higher	The system is compatible with both HDD and SSD storage, but using an SSD greatly boosts its performance together with its operating speed.
Memory (RAM)	8GB	16GB or higher	Works with 8 GB RAM, but 16 GB+ is faster as well as more effective and stable.

Processor (CPU)	Intel Core i3 1115G4	Intel Core i5 or higher (e.g., i5-1135G7 or Ryzen 5 5600U)	Both processors can run the system, but the Intel Core i5 or higher delivers a much faster processing and multitasking.
Printer	Epson Inkjet Printer	Epson EcoTank L3250 or Laser Printer	Both can print reports, but the recommended printer produces faster and more efficient prints.
Operating System	Windows 10 (64-bit)	Windows 11 Pro (64-bit)	Both operating systems are compatible, but Windows 11 Pro provides improved security and performance features.
Backup Storage	External Hard Drive 500 GB	External Hard Drive / NAS 1 TB or higher	A 500 GB drive supports basic backups, while larger storage offers more capacity for long-term data protection.
Display Monitor	19-inch L Monitor	21-inch Full HD LED Monitor	A 19-inch monitor is sufficient, but a 21-inch Full HD display provides clearer visuals and a better workspace for viewing system data.
System Unit (PC) / Laptop	Basic Desktop PC or Laptop with an entry-level spec	Mid-range Desktop PC or Laptop with higher specs	The system can run on a basic setup, but a higher specification of computer or laptop ensures better speed, multitasking, and reliability.
Uninterruptible Power Supply (UPS)	Optional	650VA or higher	The system can run without a UPS, but using one protects data and hardware from sudden power interruptions.

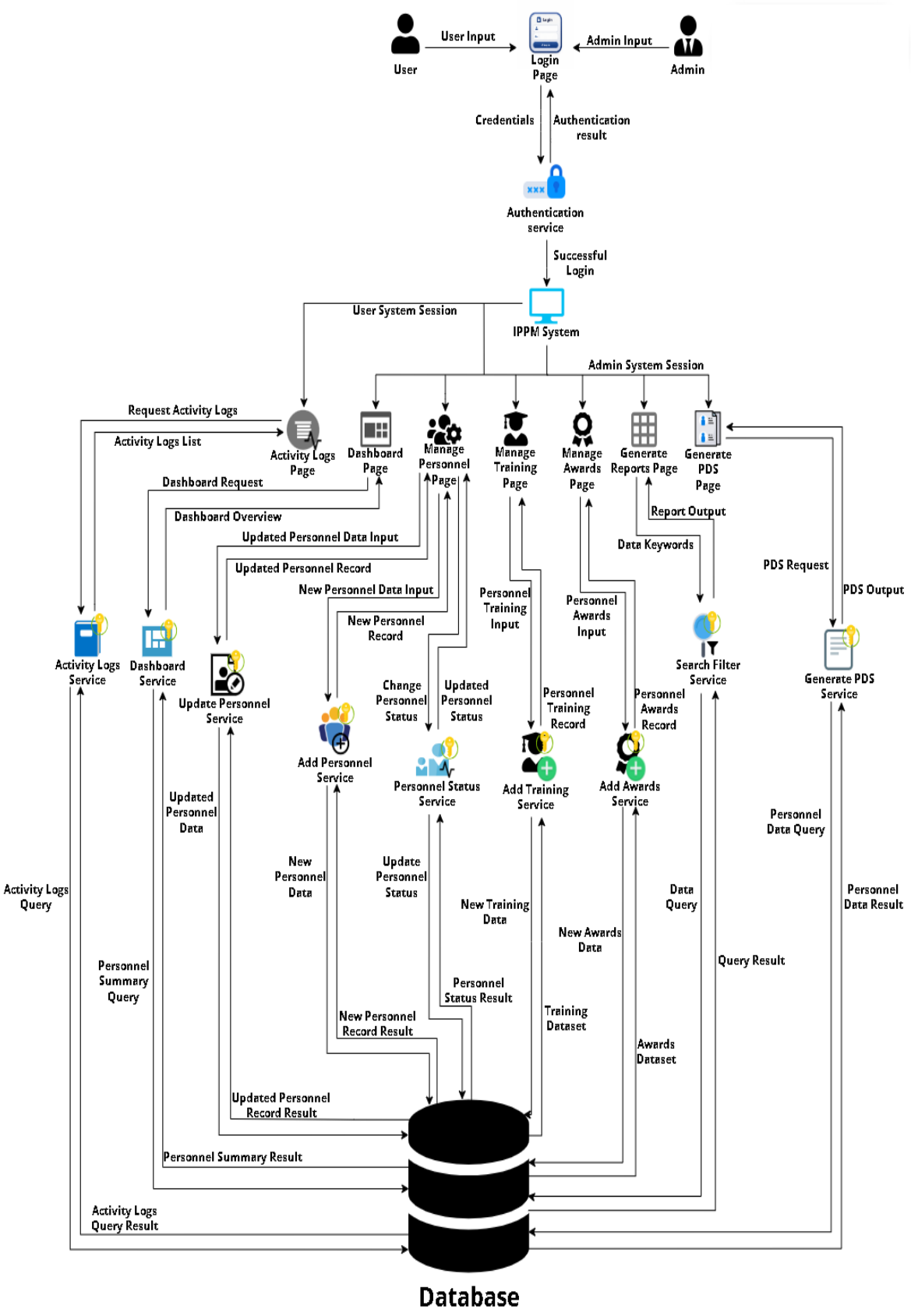


Figure 2. System Architecture Diagram

This diagram shows how users and Users log in and interact with the system to manage personnel records, statuses, trainings, awards, activity logs, and PDS reports. All actions such as adding, updating, viewing, and generating data connect to the database, which stores and provides the needed information.

Data Process Modeling

The Data Process Modelling illustrates how data flows within the Integrated Personnel Profiling and Monitoring System for the Dapitan City Police Station. It describes how information is gathered, processed, stored, and retrieved by the system. This model helps visualize the relationship between the users, processes, and the data stored in the system's database.

The proposed system follows a structured data flow, starting from data input, processing, and output generation. Personnel information, seminar records and awards are entered by the user into the system. These data are then processed and stored in the database, allowing the system to generate reports and personnel data sheets when needed.

Diagrams

In this study, diagrams serve as visual representations that illustrate how data and processes flow within the Integrated Personnel Profiling and Monitoring System for the Dapitan City Police Station. Each diagram shows how information is entered, processed, stored, and produced as system output. The visual format clarifies the structure, functions, and interactions among the users, system processes, and database. By presenting the system in a clear and organized graphical layout, each diagram simplifies complex workflows and supports easier analysis and understanding. Every illustration also provides an overview of how all modules operate together to achieve efficient personnel management.

In addition, each diagram contributes to identifying potential areas for improvement by highlighting connections, dependencies, and overall workflow behavior. The graphical representation helps developers and stakeholders visualize how every process supports the system's intended functions. Accurate documentation is also supported, ensuring proper interpretation during development, refinement, and evaluation. Overall, each diagram plays a crucial role in communicating the system's design, logic, and expected performance in a structured and easy-to-interpret format.

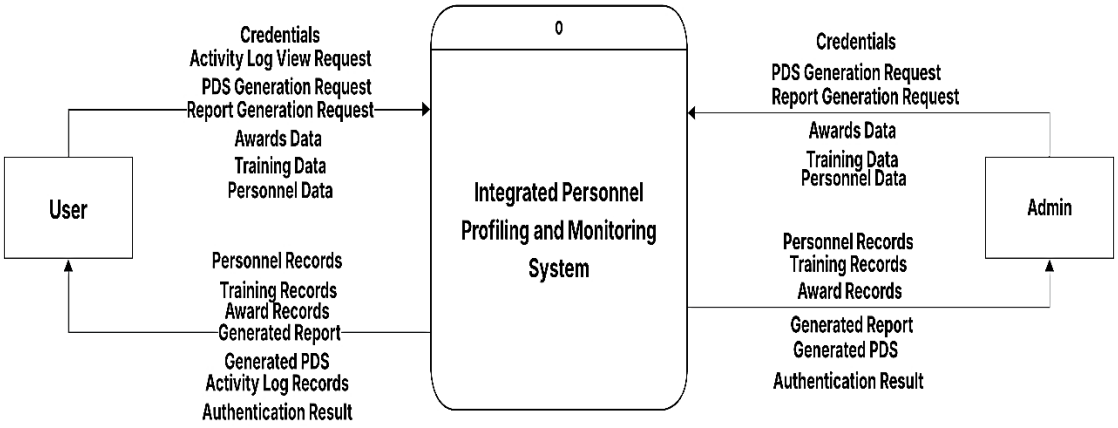


Figure 3. Context Diagram

The context diagram in Figure 2 illustrates the overall interaction between the user and the Integrated Personnel Profiling and Monitoring System. The user serves as the main user who manages all the data within the system. The user can perform several functions such as managing personnel, managing awards, and managing training activities efficiently. In return, the system produces outputs such as Personnel Data Sheets (PDS) and various reports. This diagram shows the flow of data between the user and the system, highlighting that the user is responsible for both inputting and managing information. Overall, the context diagram presents a clear overview of how the user interacts with the

system to maintain efficient, organized, and accurate personnel information management.

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates the flow of information and the major processes involved in the Integrated Personnel Profiling and Monitoring System. It shows how data moves between the user and the different system processes, including the handling of personnel records, seminars, awards, reports, and personal data sheets. Each process plays an important role in ensuring that information is properly managed, stored, and retrieved when needed. The following describes each process in detail:

1.0 Manage User Login

This process manages user access to the system by verifying login credentials and controlling authentication. When a user attempts to log in, the system validates the provided username and password against the user accounts database. If the credentials match, access is granted and the appropriate user role or permissions are loaded. If the credentials are incorrect, the system returns an error and denies entry. This process ensures secure access and protects system data by allowing only authorized users to log in.

2.0 Manage Personnel

This process is responsible for handling all activities related to personnel information. The user encodes, updates, and retrieves personnel data within the system. Information such as personnel profiles, service records, and position details are stored in the personnel database. The process ensures that all personnel information is properly maintained, easily accessible, and always up to date for administrative use. It also allows the user to change status of inactive personnel records to keep the database organized

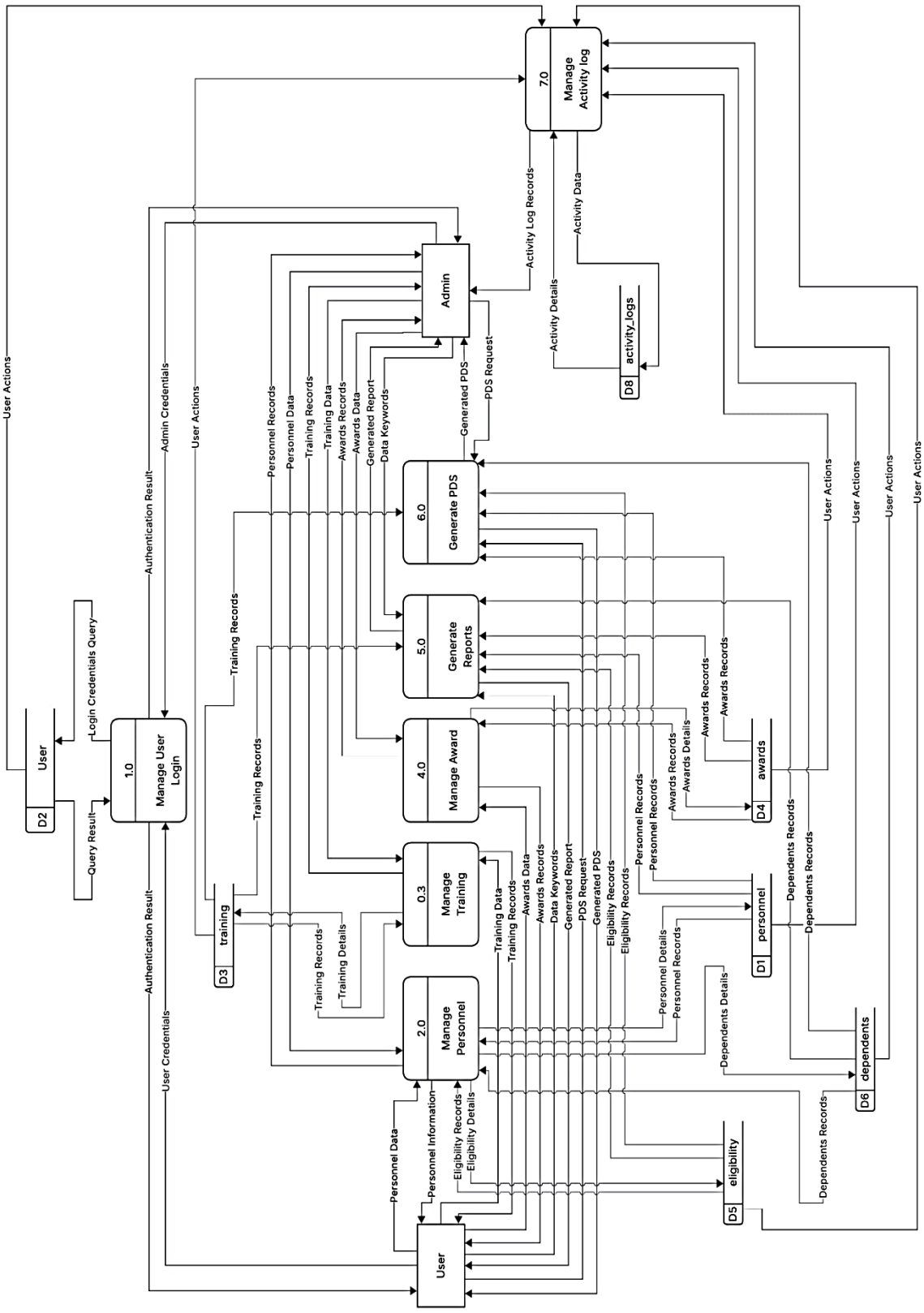


Figure 4. Data Flow Diagram

3.0 Manage Training

This process manages all training-related information of personnel. The user inputs training details such as the title, date, location, and conducting authority. The system stores this information in the training database and retrieves it upon request. Through this process, the system guarantees that all training and seminars attended by personnel are properly documented and can be accessed for evaluation, reporting, or updating personnel profiles. It supports tracking the professional growth and skill development of each personnel.

4.0 Manage Awards

The Manage Awards process handles all information regarding awards and recognitions received by personnel. The user encodes award details such as the award title, granting authority, and the date it was received. These data are stored in the awards database and can be retrieved when needed for evaluation or report generation. This process ensures that all personnel achievements are systematically documented, organized, and easily accessible for Administrative or performance evaluation purposes.

5.0 Generate PDS (Personal Data Sheet)

This process is designed to automatically generate a Personnel Data Sheet (PDS) for each personnel. When the user requests it, the system gathers all necessary information from the personnel, training aw and awards databases. The system then compiles these data into a complete and accurate PDS format. This automation minimizes manual work and ensures that all personnel data sheets are comprehensive, updated, and ready for printing or recordkeeping.

6.0 Generate Reports

The Generate Reports process is responsible for creating various reports required by the user. When a report generation request is made, the system consolidates data from

the personnel, training and awards databases to produce detailed or summary reports. These reports help the Administration in monitoring personnel performance, tracking records, and supporting decision-making activities. The process ensures that reports are accurate, consistent, and presented in an organized format for easy analysis and documentation.

7.0 Manage Activity Logs

This process records and monitors all user actions performed within the system. Every time a user logs in, updates data, performs a transaction, or accesses specific features, the system automatically captures the activity and stores it in the activity log database. The user can review these logs to track system usage, monitor user behavior, and ensure accountability. This process helps maintain system integrity by providing a reliable record of actions that can be used for auditing, security checks, and troubleshooting.

Use Case Diagram

The Use Case model of the system presents a clear visualization of how the user interacts with the system's main functions and features. It identifies the primary actor, which is the user, along with the various use cases or processes the user is allowed to perform within the system. These use cases outline the specific actions, such as managing personnel information, updating records, and generating reports, that contribute to the system's overall functionality. The object model ensures that all necessary functions are clearly defined, organized, and accessible only to authorized users to maintain security and accuracy. It also serves as a blueprint for system design, making it easier to implement, test, and maintain the system. By providing a structured representation, the model minimizes potential errors and ensures that all user requirements are addressed. Overall, it enhances both the clarity and efficiency of system development and management.

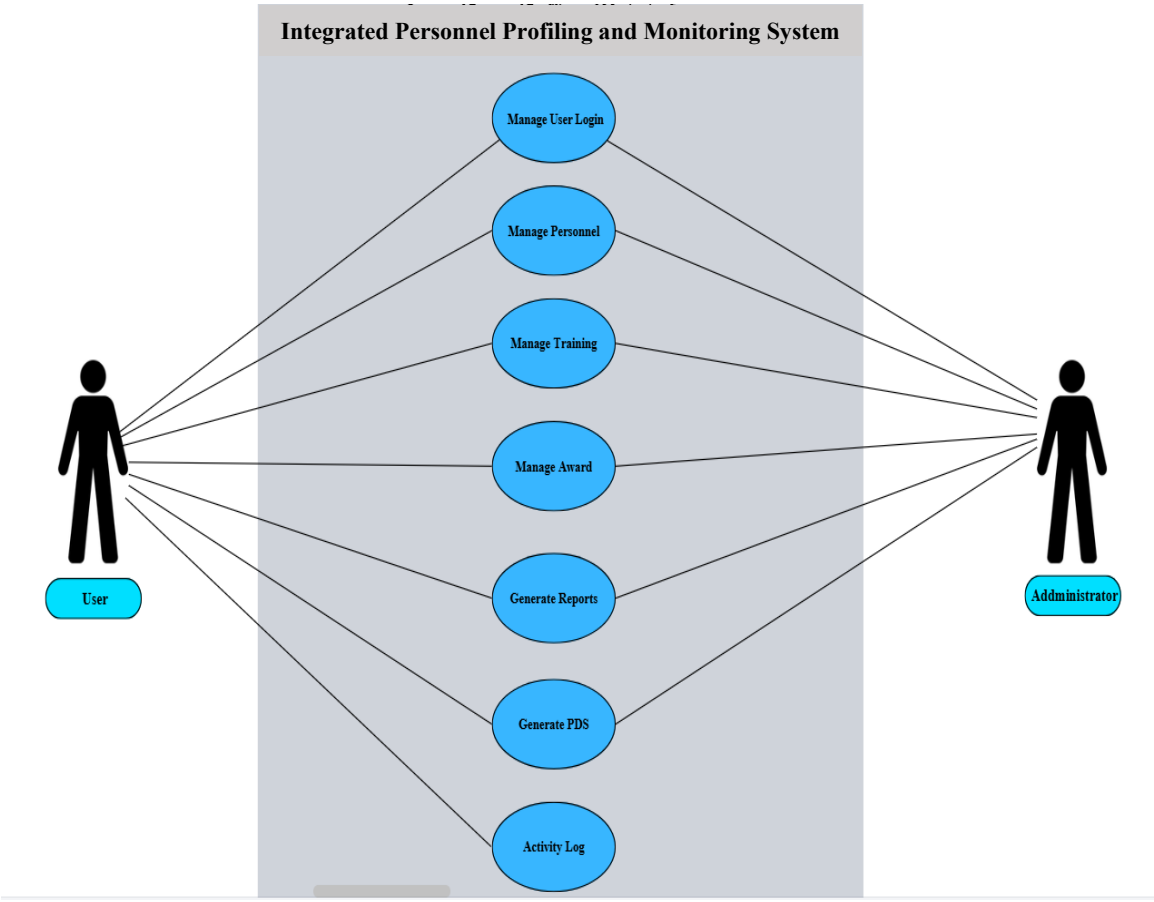


Figure 5. Use Case Diagram

As shown in Figure 5, the user is the primary actor who has full control over all system operations and serves as the central point of interaction within the platform. The user is responsible for managing personnel records, archiving data for future reference, recording seminars or training, handling awards, generating Personal Data Sheets (PDS), producing various reports as required, and tracks all the users activities. Each of these processes represents a key function of the system, ensuring that personnel information is accurately maintained, updated, and easily accessible. These tasks also demonstrate how the system supports efficient documentation, organized data management, and streamlined workflow for administrative activities. Overall, this diagram emphasizes the user's crucial role in operating and maintaining the integrity of the entire system.

1. Manage User Login

This function verifies user credentials to ensure that only authorized individuals can access the system. It helps maintain system security by checking usernames and passwords before granting entry. Through this process, sensitive personnel information is protected from unauthorized use.

2. Manage Personnel

This function allows the User to add, update and view personnel records. It ensures that all personnel information is accurate and up to date.

3. Manage Training

The user can record and manage all seminar or training activities attended by personnel, ensuring that each training experience is properly documented. They can input important details such as the course title, duration, location, and certificate information maintain accurate and complete records. This process helps the system track personnel development and supports better planning for future training needs.

4. Manage Awards

This function allows the User to input and organize all awards or recognitions received by personnel, including the type of award, issuing authority, and date received.

5. Generate PDS (Personal Data Sheet)

The User can automatically generate a PDS that summarizes a personnel's information, including personal details, service record, and other relevant data.

6. Generate Reports

The system can produce various reports upon request, such as personnel listings, birthdays, marital status, eligibility, and other important summaries needed for administra-

tive use.

7. Manage Activity Logs

This function records all user actions within the system, helping monitor activities and ensure accountability. It logs key events such as login attempts, data updates, and record changes to support transparency, security, and reliable audit trails. These logs make it easier to trace actions, resolve issues, and maintain the integrity of system operations. The Use Case diagram provides a clear view of how users interact with the system's main features, showing how personnel management, archiving, and report generation work together. It helps explain user roles, system behavior, and overall functionality, making the workflow easier to understand and aiding developers in implementing and maintaining the system. This ensures that both users and developers gain a complete understanding of how each component connects.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) of the Integrated Personnel Profiling and Monitoring System shows the overall structure of the database and how the data in the system are connected to each other. It helps in organizing, storing, and retrieving information efficiently. The ERD is important because it guides the proper design of the database and prevents duplication or loss of data.

1. User Entity

The User table manages the system login accounts assigned to organization users who operate the system. It stores authentication-related information such as username, password, and email. It also includes fields that support account recovery, such as reset codes and expiration timestamps. Each user is uniquely identified by a Primary Key

(user_id). This entity is crucial for system security and access control. It ensures that only authorized individuals can access specific system functions, protecting sensitive personnel data.

2. Personnel Entity

The Personnel table is the core of the system. It stores all essential information about each personnel, including their personal details such as rank, family name, first name, middle name, birthdate, blood type, and marital status. It also includes employment information such as address, atm account number, date of appointment, eligibility, schooling, and contact details. Additionally, it keeps physical and emergency information such as height, weight, emergency contact name, emergency contact number, and emergencyaddress. Each personnel record is uniquely identified by a Primary Key (personnel_id)**.**

3. Activity Logs Entity

The Activity Logs table records all significant system interactions performed by a user. It includes the timestamp, the page accessed, the action executed, and other relevant details such as status and a short description of the event. The table maintains foreign key relationships linking logs to both the User and Personnel entities, allowing user to audit usage behavior and trace operational history. Each log entry is uniquely identified by log_id.

4. Awards Entity

The Awards table stores information regarding recognitions and honors received by personnel during their service. It contains details such as the nature of the award, the authority that granted it, the award date, and any additional remarks. Each award record is

associated with a personnel member through a foreign key, establishing a one-to-many relationship where a personnel may have multiple awards. The primary key for this entity is award_id.

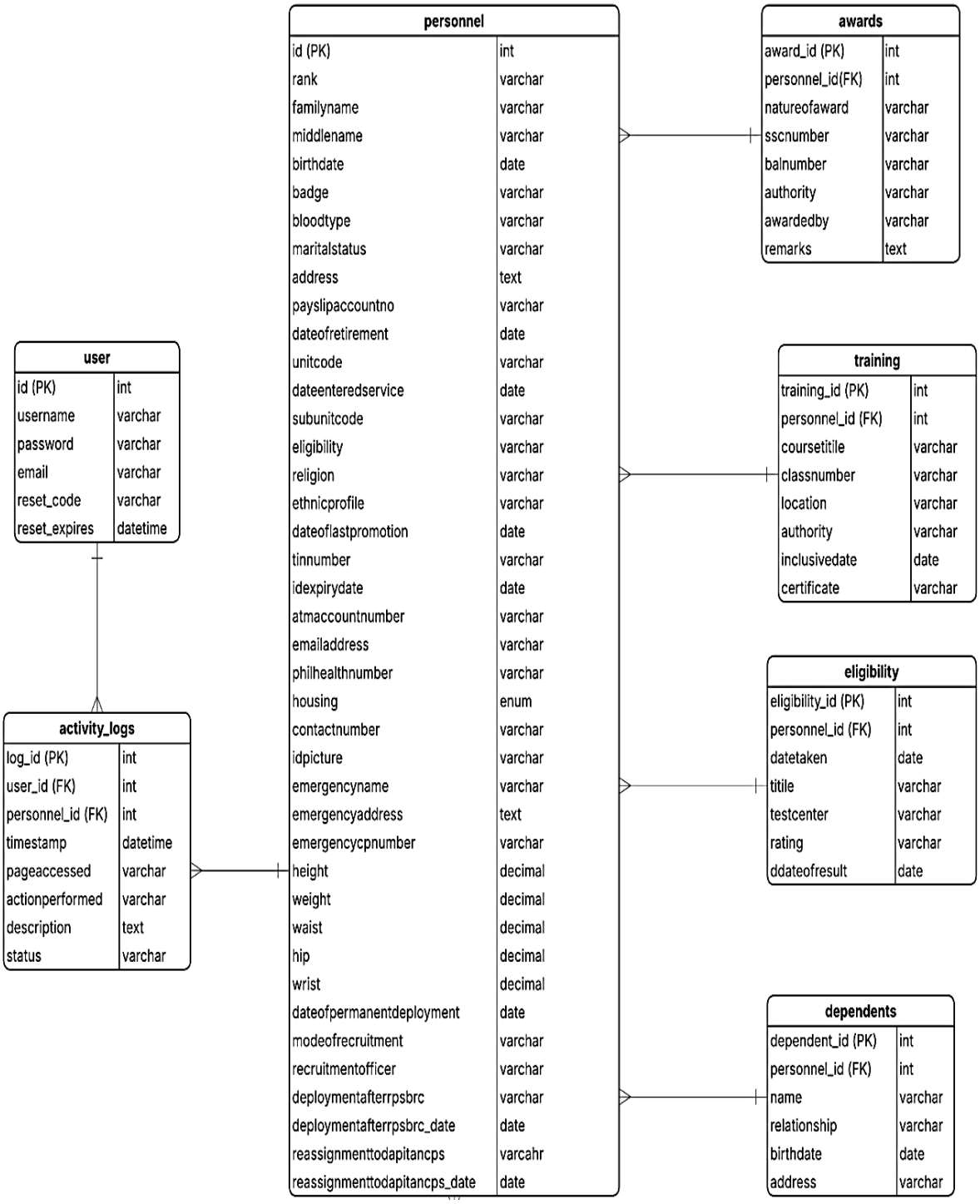


Figure 6. Entity Relationship Diagram

5. Training Entity

The Training table tracks professional courses, seminars, and skill-development activities attended by personnel. It includes information such as course title, location, authority conducting the training, and certificate details. The inclusive dates of the training are also recorded. Each training entry is linked to a specific personnel via a foreign key. The primary key for the table is training_id, ensuring each training record is uniquely identifiable. This structure helps maintain accurate records of personnel development and career progression. It also supports generating reports that summarize training accomplishments for evaluation or administrative purposes.

6. Eligibility Entity

The Eligibility table captures the career eligibility credentials of personnel, such as government-mandated exams or certifications. It stores related information including title, test center, rating, date of exam, and the date the result was released. Through the foreign key personnel_id, multiple eligibility records can be assigned to one personnel. Each record is uniquely defined by the eligibility_id primary key.

7. Dependents Entity

The Dependents table contains information about individuals who are directly dependent on the personnel, such as children or spouses. It stores their basic details including name, birthdate, relationship to the personnel, and address. This helps in managing personnel benefits and personal records. Each dependent is linked to specific personnel through a foreign key, and uniquely identified by dependent_id.

Personnel to Awards. This relationship is one-to-many because a single personnel member may receive multiple awards throughout their service, but each award can only belong to one personnel record.

Personnel to Training. This is a one-to-many relationship since one personnel member can attend various training programs over the course of their career, while each training record is linked to only one personnel.

Personnel to Eligibility. This relationship is one-to-many because a personnel member may acquire several eligibility certifications or pass multiple examinations, but each eligibility record is tied to only one personnel. This structure ensures that all eligibility details are properly organized and accurately traced back to the correct individual. It also facilitates efficient reporting and retrieval of eligibility information for each personnel member.

Personnel to Dependents. This is a one-to-many relationship as a personnel member can have multiple dependents such as their spouse, children, or parents, while each dependent record corresponds to only one personnel.

Personnel to Activity Logs. This is a one-to-many relationship because a personnel member may have multiple recorded interactions or actions logged within the system, yet each log entry is linked to only one personnel.

User to Activity Logs. This relationship is one-to-many because a single user account may perform several activities while using the system, and every recorded log entry is associated with only one user.

Overall Description

This Entity Relationship Diagram (ERD) presents the organized structure of how

personnel information and related records are managed within the system. Personnel data serves as the main component and is supported by linked information stored in the Awards, Training, Eligibility, Dependents, PDF Library, and Activity Logs tables. These relationships ensure that all relevant data connected to each personnel record is properly tracked and maintained. The User entity oversees system access and operations, ensuring security, accuracy, and efficient management of all personnel information stored in the database

Data Dictionary

The Data Dictionary provides detailed information about all the data elements used in the system. It defines each table, its fields, data types, lengths, keys, and a short description of their purpose. This ensures that data is structured consistently, relationships are clearly identified, and database operations are properly managed.

Each table in the database represents a specific entity within the system. The following are the detailed descriptions of all tables and their corresponding fields, providing clearer understanding and structure.

Table 6: user

The user table stores account information for system user. It contains details necessary for managing access control and password recovery functions.

Field Name	Field Type	Length	Key	Description
Id	int	11	PK	The unique number for each User record.
username	varchar	50		The User's login name.
password	varchar	255		The User's password used to log in.
Email	varchar	100		The User's email address.

reset_code	varchar	10		A short code used to reset the User's password.
reset_expires	datetime	-		The date and time when the reset code will expire.

Table 7: awards

The Awards table keeps records of all awards and recognitions received by active personnel.

Field Name	Field Type	Length	Key	Description
award_id	int	11	PK	The unique ID of each award record.
personnel_id	int	11	FK	The ID of the personnel who received the award.
natureofaward	varchar	255		The type or name of the award.
sscnumber	varchar	100		The SSC reference number related to the award.
balnumber	varchar	100		The BAL reference number.
authority	varchar	255		The office or department that approved the award.
awardedby	varchar	255		The person or group who gave the award.
remarks	text			Notes or comments about the award.

Table 8: personnel

The Personnel table serves as the main table of the database. It stores all active personnel records, including personal, employment, and physical information.

Field Name	Field Type	Length	Key	Description
Id	int	11	PK	The unique number of each personnel record.
Rank	varchar	50		The position or rank of the personnel.
familyname	varchar	50		The last name of the personnel.
middlename	varchar	50		The middle name of the personnel.
birthdate	date	-		The birth date of the personnel.
Badge	varchar	50		The personnel's badge number.
bloodtype	varchar	5		The blood type of the personnel.
maritalstatus	varchar	50		The marital or civil status (single, married, etc.).
address	varchar	20		The current home address of the personnel.
payslipaccountno	varchar	50		The account number for payroll or payslip.
dateofretirement	date			The date wherein a certain personnel will retire.
unitcode	varchar	20		The code of the unit where the personnel is assigned.
dateenteredservice	date	-		The date when personnel started service.
subunitcode	varchar	20		The code of the subunit where personnel belong.

eligibility	varchar	50		The eligibility of a certain personnel.
religion	varchar	50		The religion of a certain personnel.
ethnicprofile	varchar	50		The ethnic background of the personnel.
dateoflastpromotion	date	-		The date where a certain personnel was lastly being promoted.
tinnumber	varchar	20		The TIN number of the personnel.
schooling	varchar	50		The school or training institution attended.
idnumber	varchar	50		The ID number of the personnel.
educationalattainmentcourse	varchar	100		The educational attained by the personnel or course taken.
idexpirydate	date	-		The date when the personnel's ID will expire.
atmaccountnumber	varchar	50		The ATM account number for salary.
emailaddress	varchar	100		The email address of the personnel.
philhealthnumber	varchar	20		The PhilHealth number of the personnel.
housing` enum('With House and Lot','With Lot Only','With House Only','Living With Parents/Relatives')				The housing type (e.g., With House, With Lot, Living With Parents).

contactnumber	varchar	20		The contact number of the personnel.
idpicture	varchar	255		The ID picture or photo file.
emergencyaddress	text			The emergency contact address.
emergencycpnumber	varchar	L20		The emergency contact number.
Height	decimal	5,2		The height of the personnel in meters.
weight	decimal	5,2		The weight of personnel in kilograms.
Waist	decimal	5,2		The waist measurement.
Hip	decimal	5,2		The hip measurement.
Wrist	decimal	5,2		The wrist measurement.
dateofpermanentappointment	date	-		The date when personnel became permanent in their official service records.
modeofrecruitment	varchar	50		The way the personnel was recruited.
recruitmentofficer	varchar	100		The name of the officer who recruited a personnel.
deploymentafterpsbrc	varchar	100		The first assignment after training (PSBRC).
deploymentafterpsbrc_date	date	-		The date of deployment.
reassignmenttodapitancps	varchar	100		The next reassignment location.
reassignmenttodapitancps_date	date	-		The date when reassigned to new post.

Table 9: Training

The Training table stores all records of training courses attended by active personnel. It includes important details such as course titles, dates, locations, and related certifications. This table helps ensure that all training information is properly documented for tracking personnel development and validating qualifications.

Field Name	Field Type	Length	Key	Description
training_id	int	11	PK	The unique ID of each training record.
personnel_id	int	11	FK	The ID of the personnel who attended the training.
coursetitle	varchar	255		The name or title of the training course.
classnumber	varchar	100		The class number of the training.
location	varchar	255		The place where the training was held.
authority	varchar	255		The office or person who authorized the training.
inclusivedate	date	-		The start and end dates of the training.

Table 10: eligibility

The eligibility table stores all records of eligibility acquired by active personnel, ensuring that each individual's qualifications are accurately tracked and easily accessible.

Field Name	Field Type	Length	Key	Description
eligibility_id	int	11	PK	The unique ID of each training record.
personnel_id	int	11	FK	The ID of the personnel who attended the training.

datetaken	varchar	255		The name or title of the training course.
Title	varchar	100		The class number of the training.
testcenter	varchar	255		The place where the training was held.
Rating	varchar	255		The office or person who authorized the training.
dateofresult	date	-		The start and end dates of the training.

Table 11: dependents

The dependents table stores information about all dependents of active personnel, including basic details to support personnel records and related processes.

Field Name	Field Type	Length	Key	Description
dependent_id	int	11	PK	The unique ID of each training record.
personnel_id	int	11	FK	The ID of the personnel who attended the training.
Name	varchar	255		The name or title of the training course.
relationship	varchar	100		The class number of the training.
birthdate	varchar	255		The place where the training was held.
address	varchar	255		The office or person who authorized the training.

Table 12: activity_logs

The activity logs table stores records of all actions and activities performed within the system, tracking user interactions and system events for monitoring and auditing purposes.

Field Name	Field Type	Length	Key	Description
log_id	int	11	PK	The unique ID of each activity log record.
user_id	int	11	FK	The ID of the user who performed the action.
personnel_id	int	11		The ID of the personnel related to the logged activity, if applicable.
timestamp	datetime	255		The date and time when the activity occurred.
pageaccessed	varchar	255		The pages and modules accessed by the user.
actionperformed	varchar	255		The specific action or operation executed by the user.
description	text	255		Additional details about the performed action.
Status	varchar	255		The result or outcome of the action taken.

Forms

The Integrated Personnel Profiling and Monitoring System (IPPM) contains several forms that serve as the main interface for managing personnel information. These forms allow users or user to input, view, update, and monitor personnel data efficiently. They are designed to be intuitive, ensuring ease of use for all levels of staff. Each form supports data validation to maintain accuracy and prevent errors. Additionally, the forms provide quick access to historical records and generate reports to aid in decision-making.

Prototype

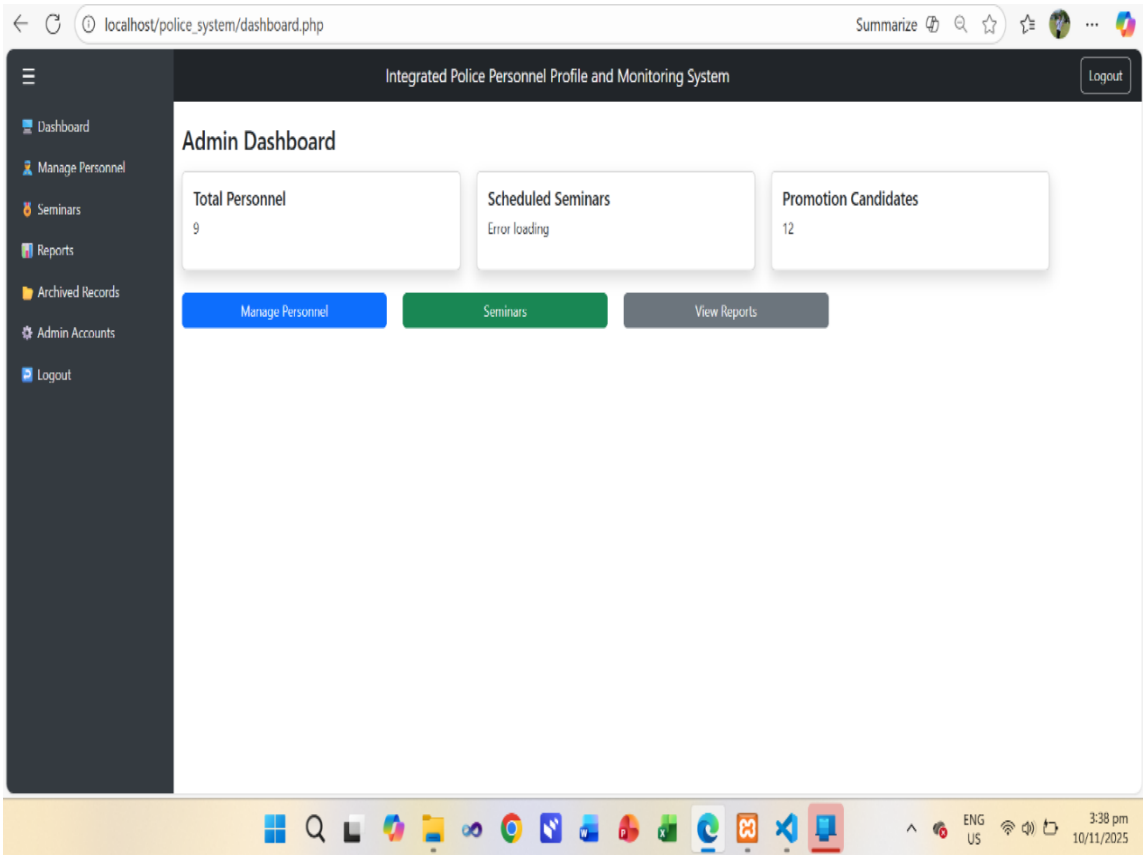


Figure 7. User Dashboard Form Prototype

This figure shows the dashboard's initial design, highlighting its main interface, navigation features, and key user functions for managing records and activities.

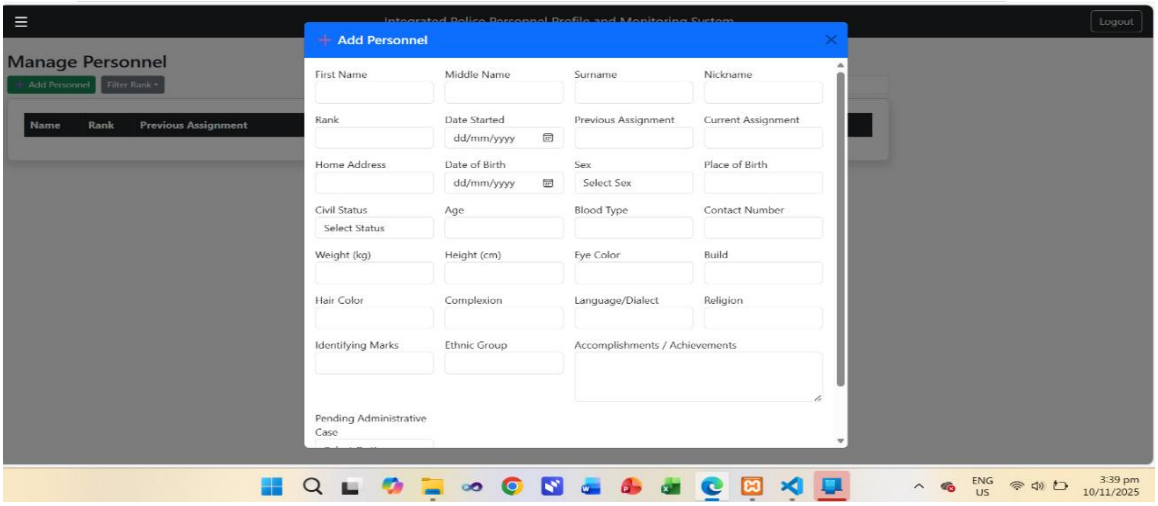


Figure 8. Add Personnel Form Prototype

This figure shows the "Add Personnel" form prototype, allowing users to input new personnel details like name, contact, position, and department. It is user-friendly and includes validation checks to ensure accurate data entry.

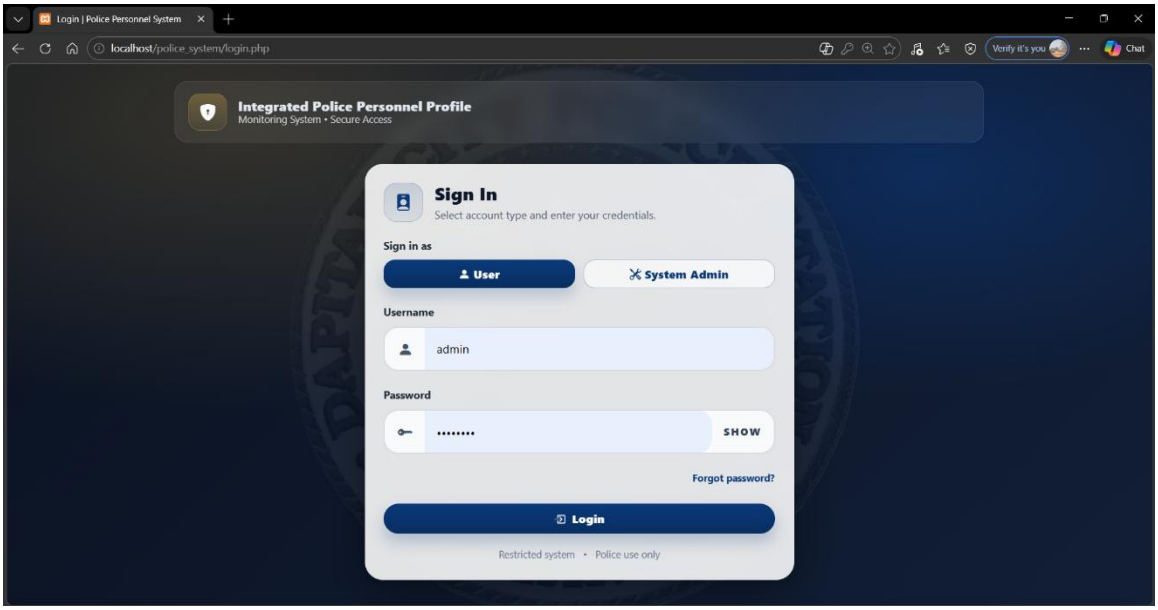


Figure 9. Login page for the IPPM System

This form is where the User logs in to access the Integrated Personnel Profiling and Monitoring System.

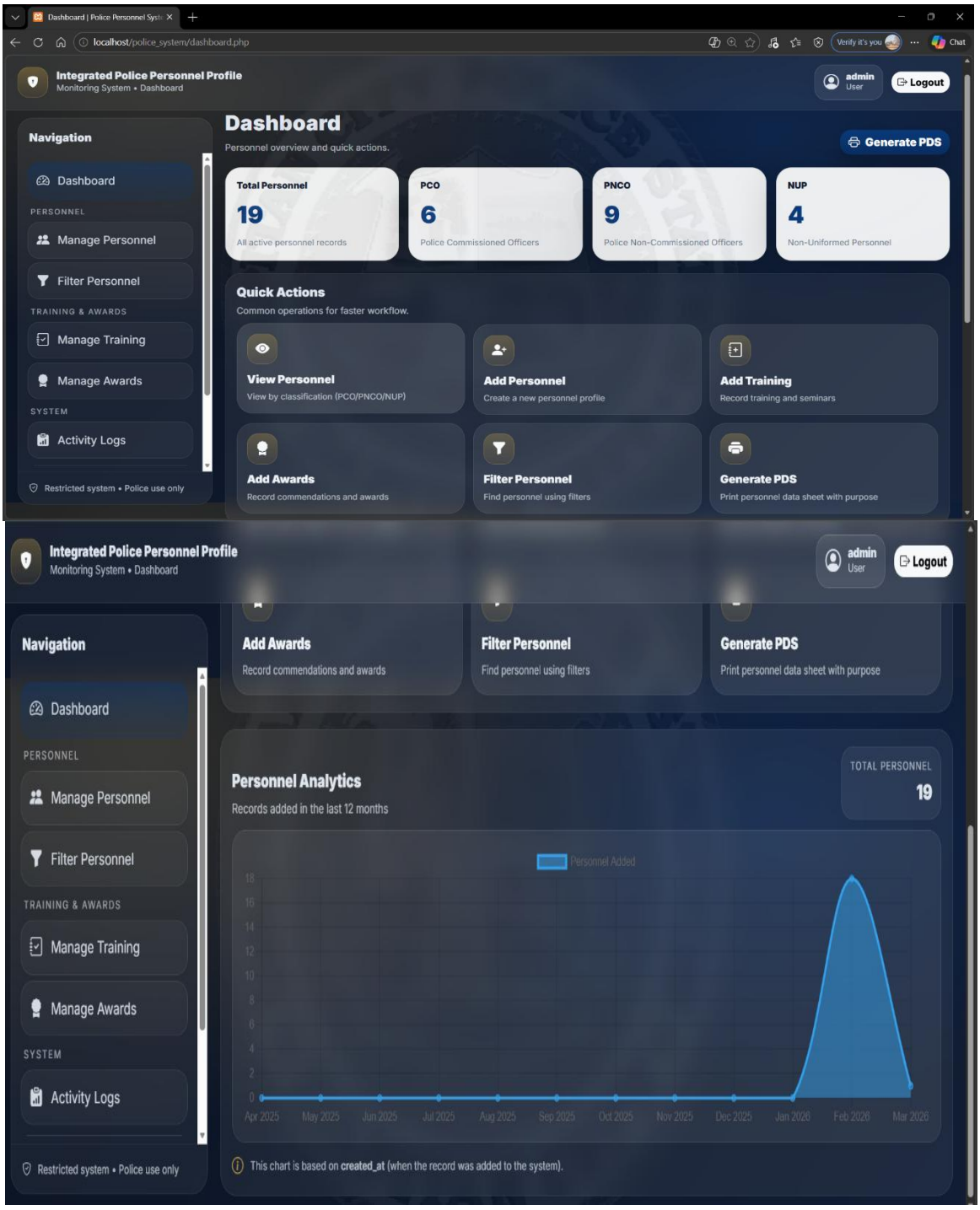


Figure 10. Dashboard for the IPPM System

Displays the total number of personnel and provides quick access buttons for viewing, adding, filtering, and generating records.

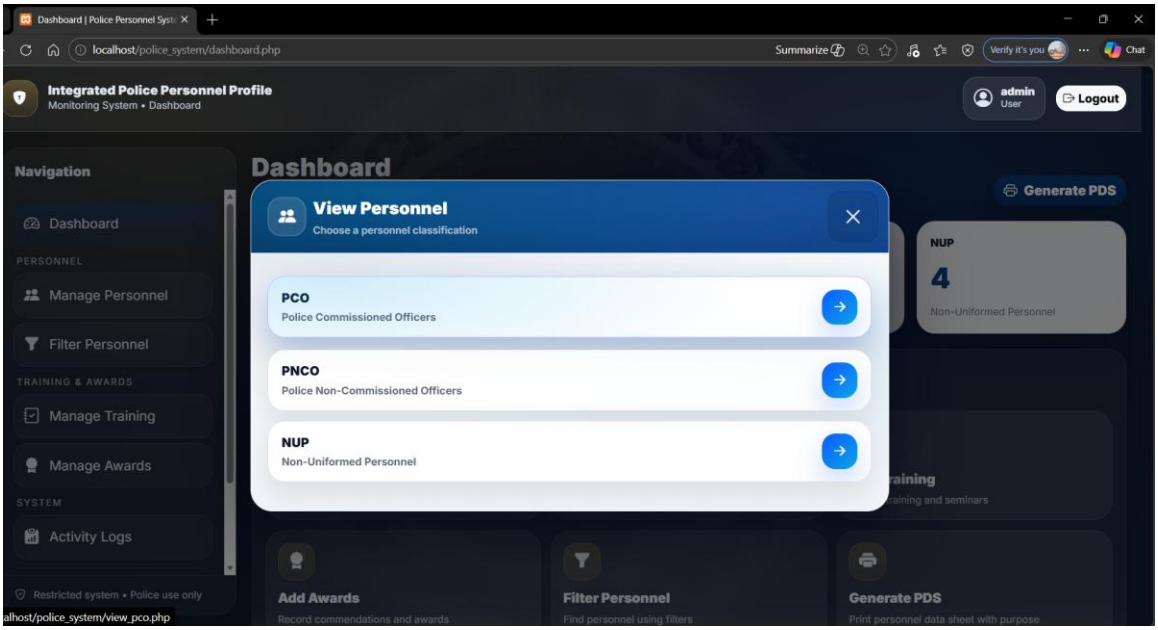


Figure 11. View Personnel Window

Appears when the User clicks the "View Personnel" button, allowing selection of a personnel type.

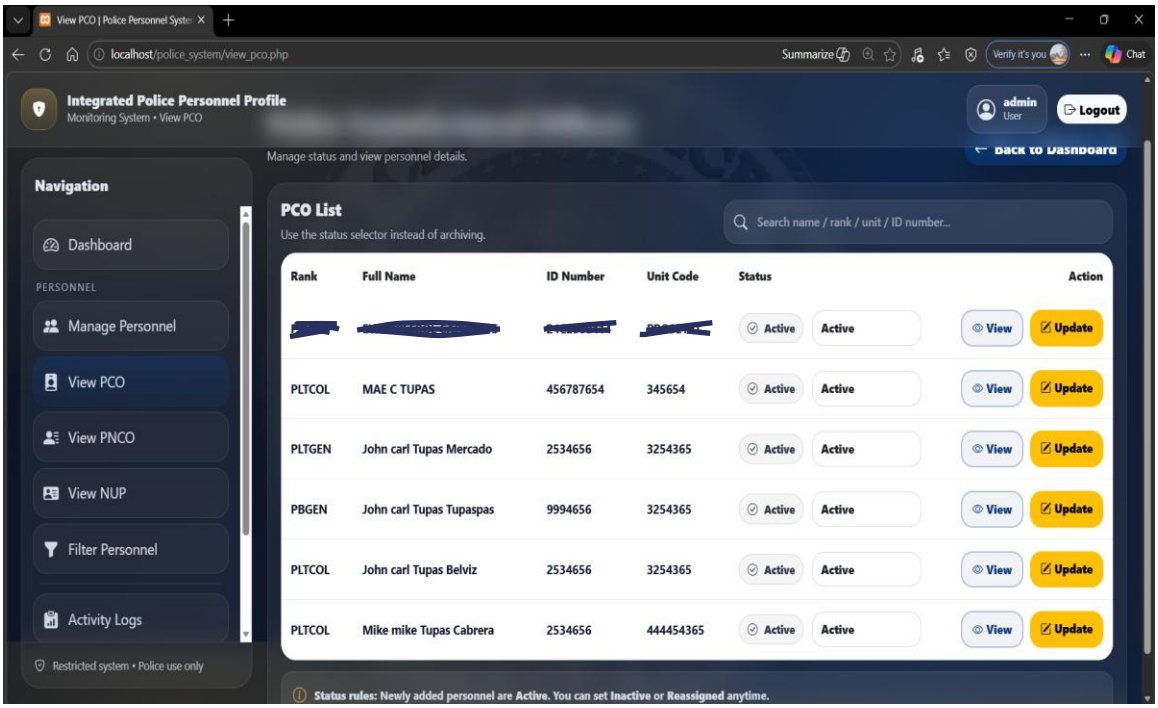


Figure 12. Personnel List Form

Shows the list of all personnel with options to view or edit their records.



Figure 15. Add Personnel Form.

Used to add new personnel information to the system.

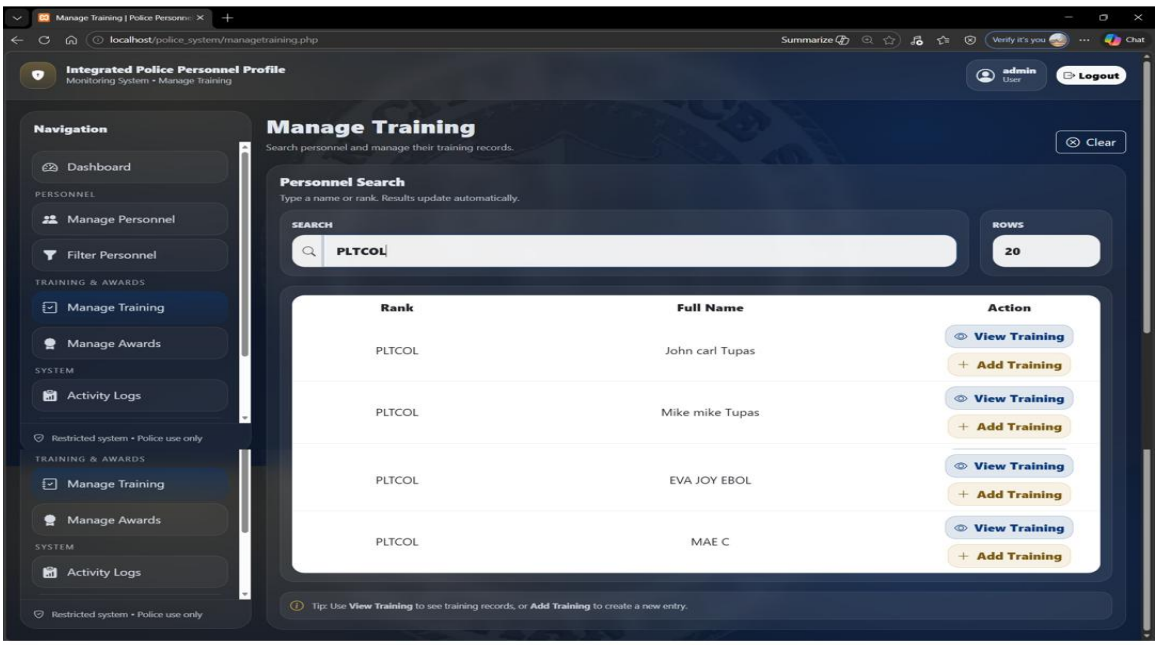


Figure 16. Manage Training Form

Used to view and add personnel training records.

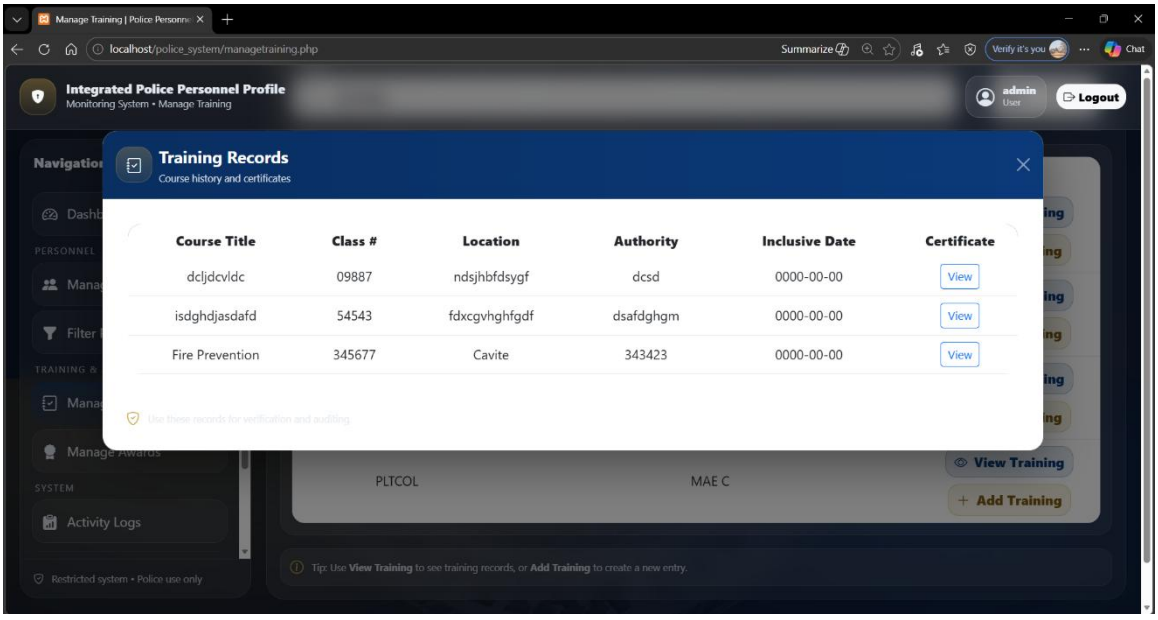


Figure 17. Personnel's Individual Training List

Shows all training taken by a specific personnel.

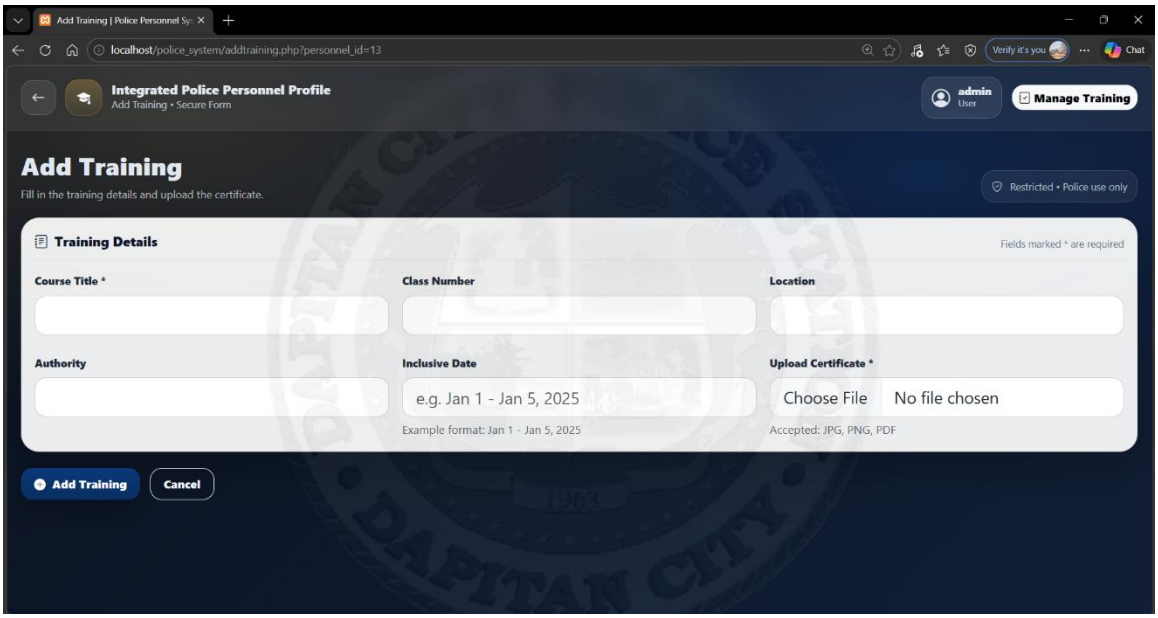


Figure 18. Add Training Form

Used to record new training details and upload certificates.

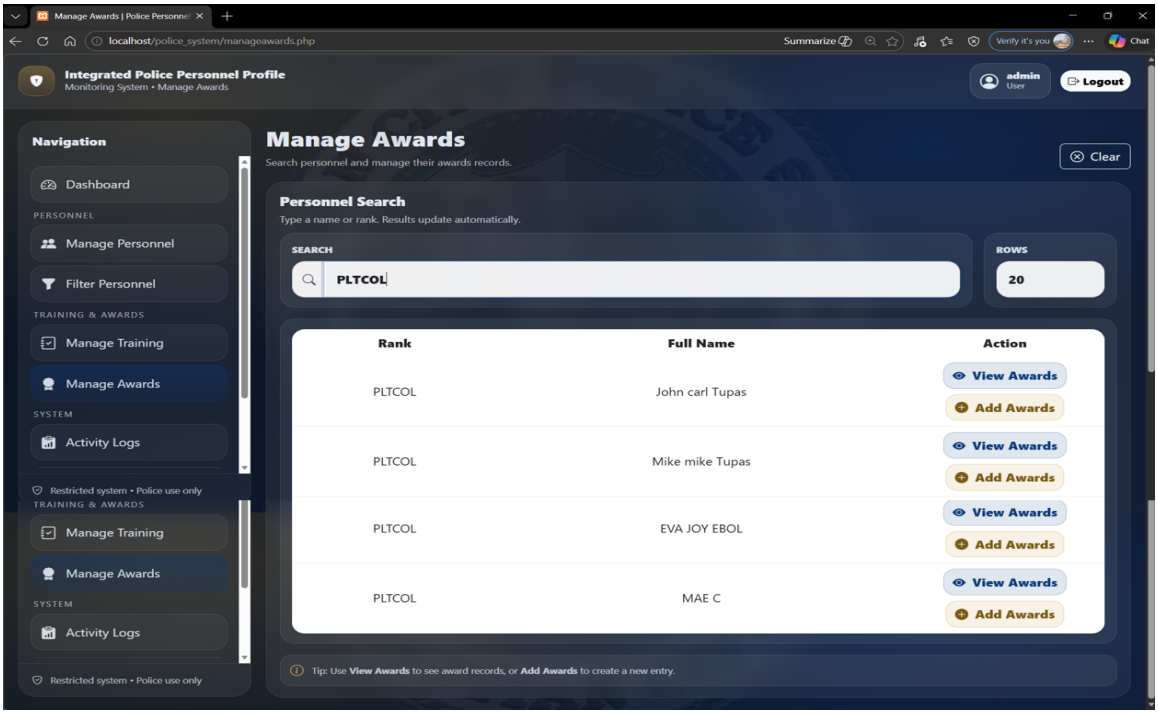


Figure 19. Manage Awards Form

Used to view and add personnel award records.

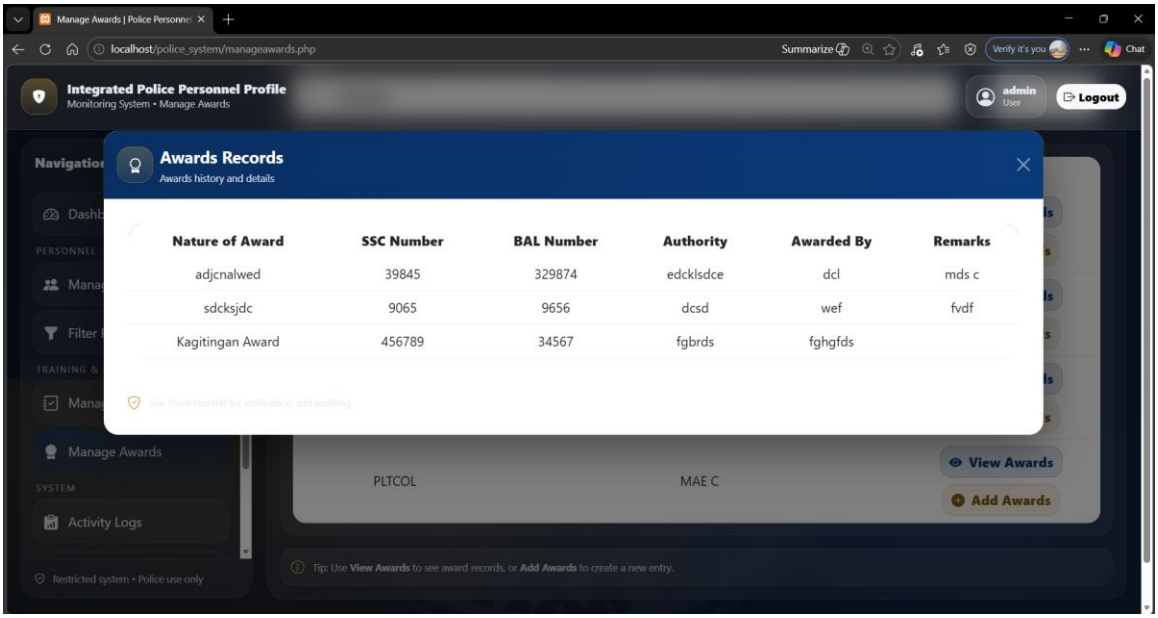


Figure 20. Personnel's Individual Awards List

Shows all training taken by specific personnel.

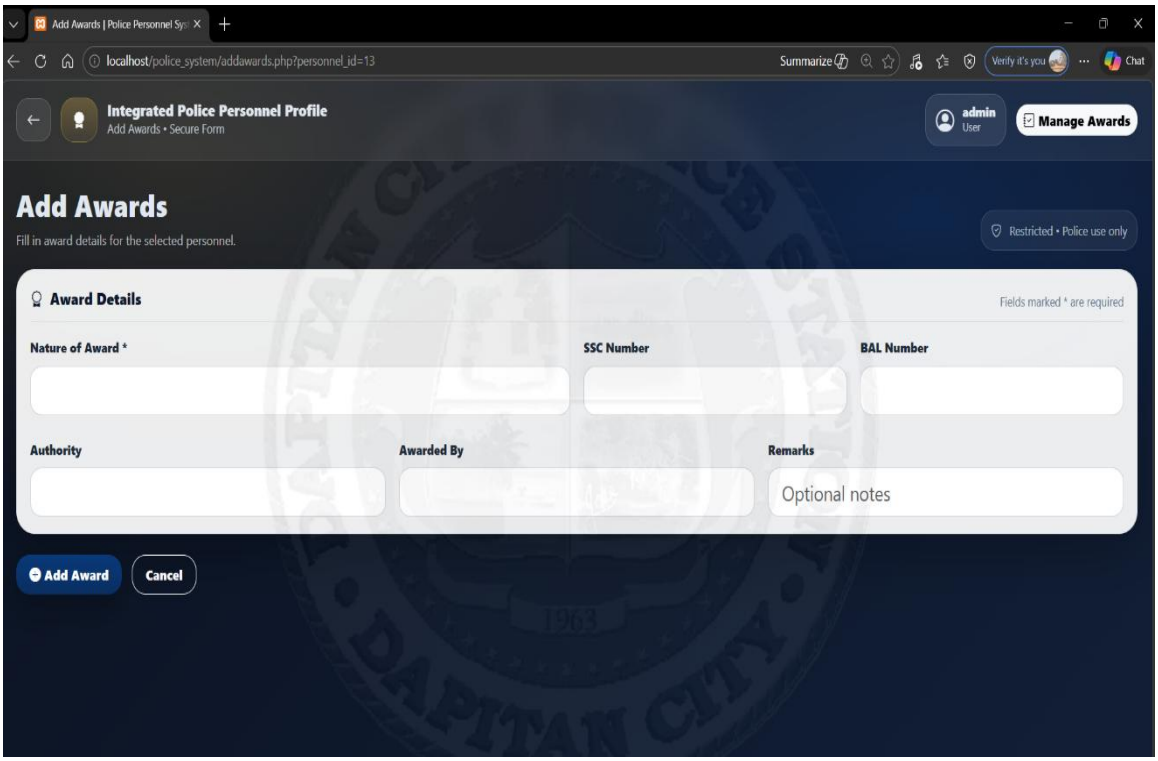


Figure 21. Add Awards Form

Used to record new Awards details

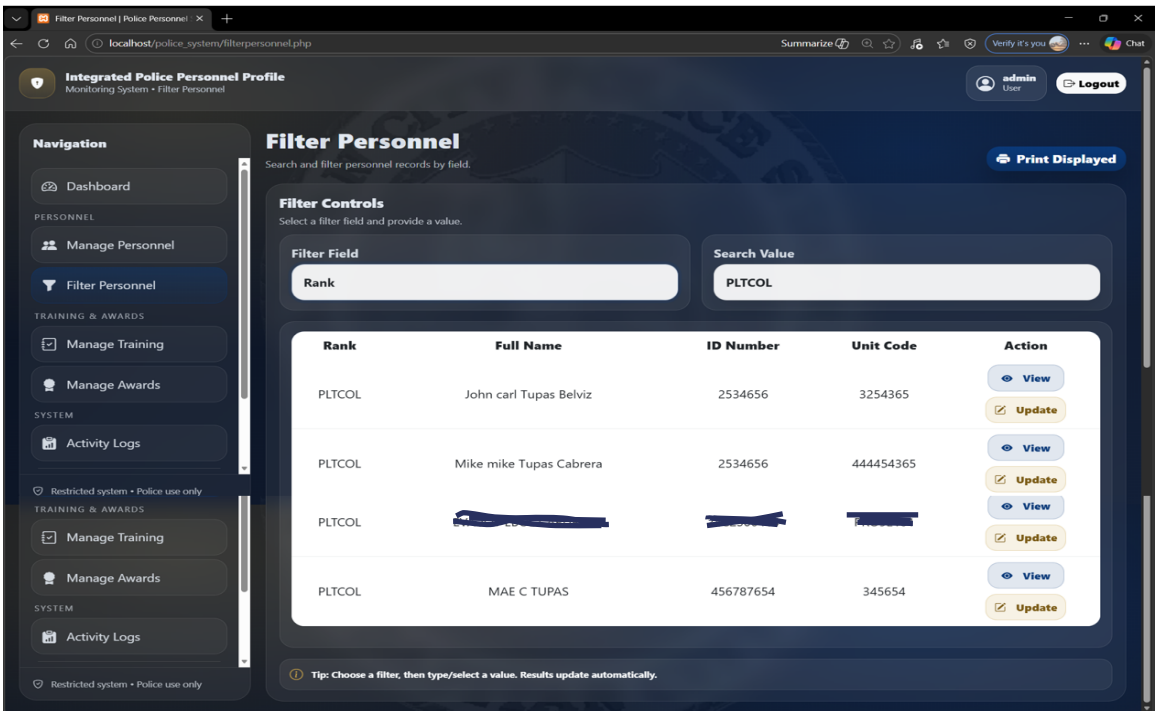


Figure 22. Filter Personnel Form

Used to search and filter personnel easily.

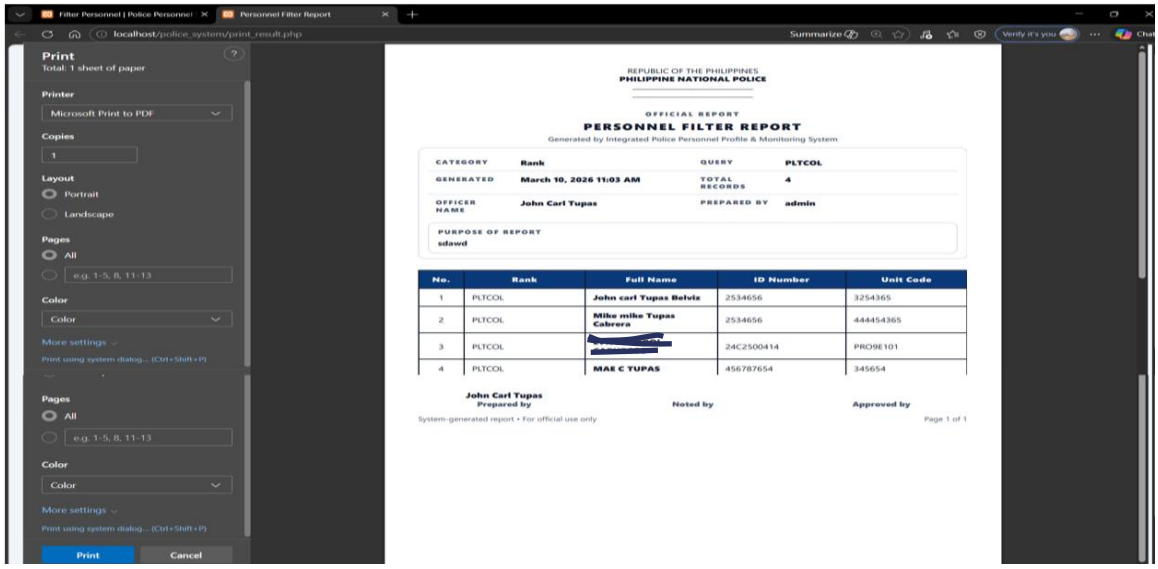


Figure 23. Filtered Personnel List Printable

Shows the list of filtered personnel ready for printing

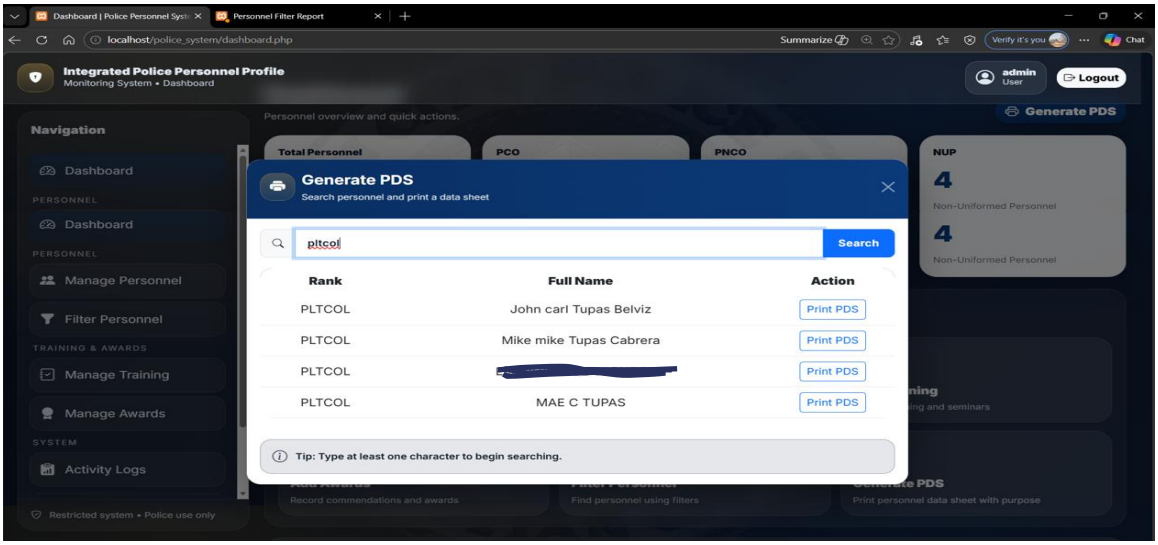


Figure 24. Generate PDS Window

Used to select and generate a personnel's data sheet (PDS).

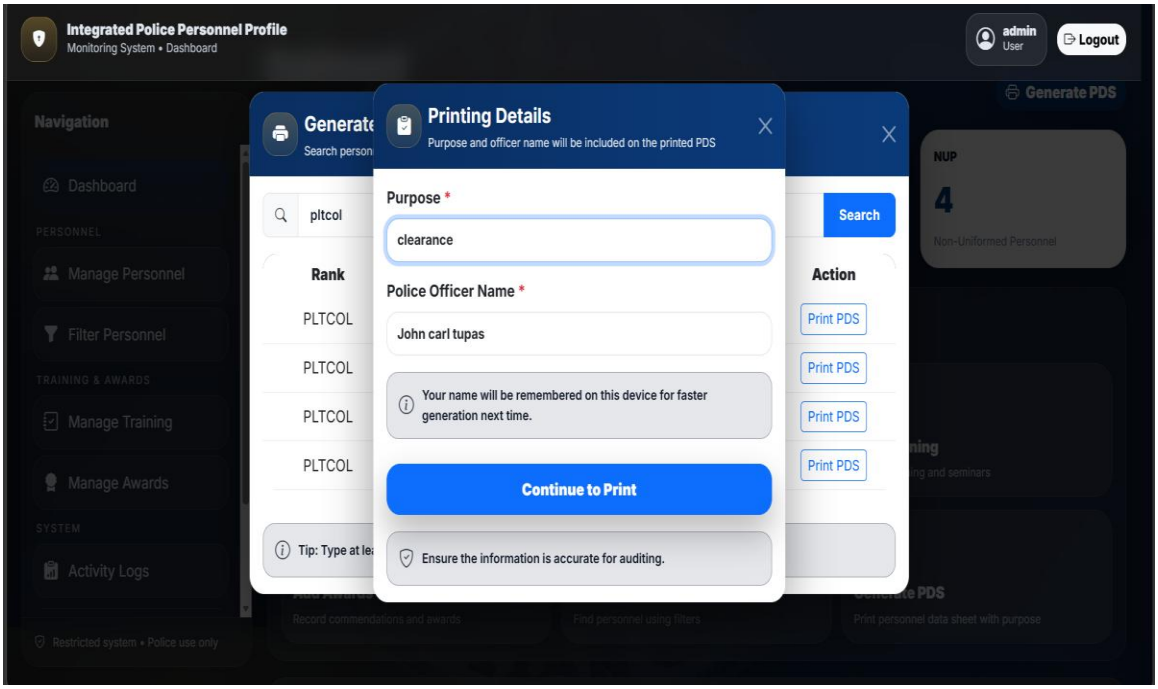


Figure 25. PDS Purpose Window

Displays a modal where the user enters the purpose before printing the PDS.

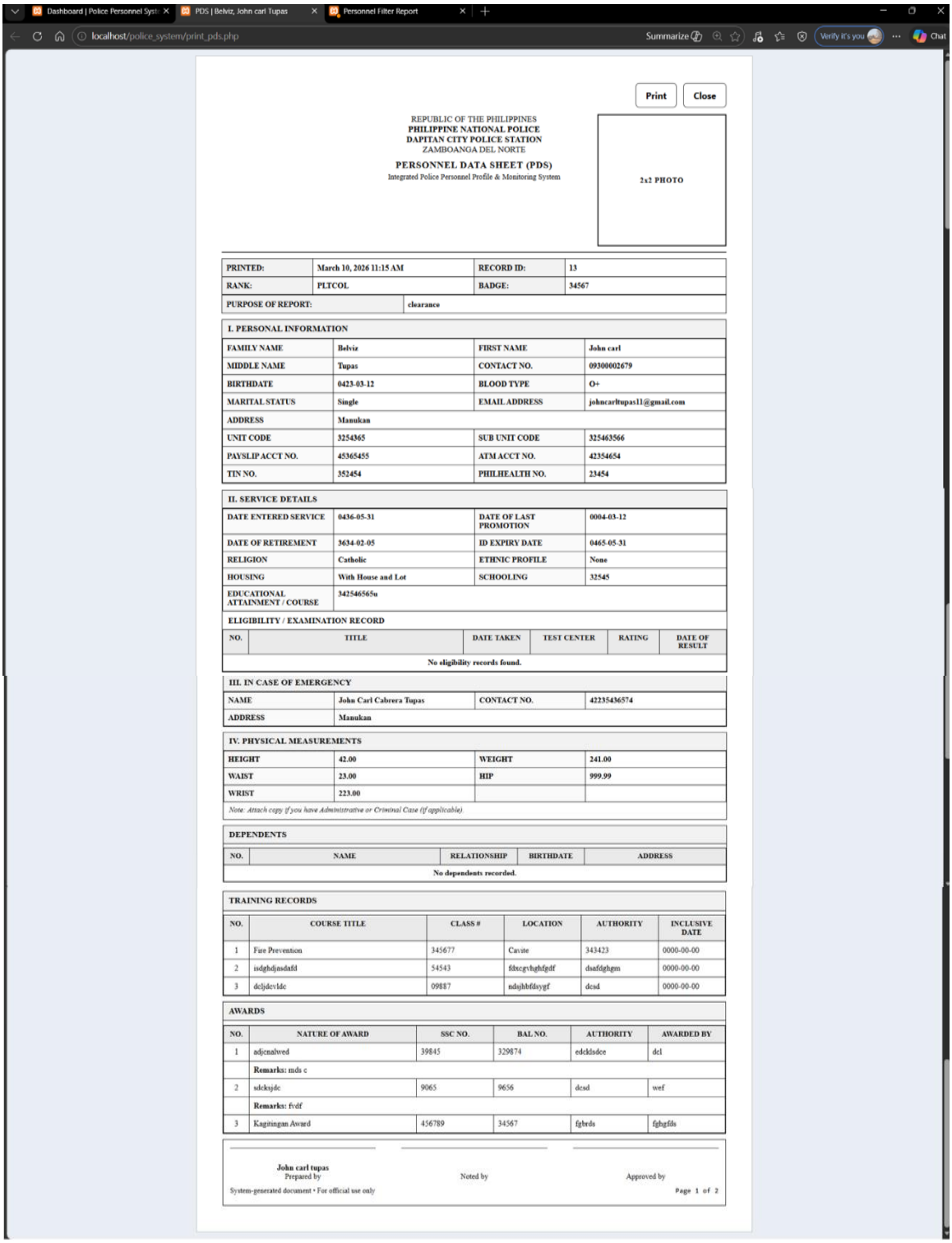


Figure 26. PDS Printable with Purpose Watermark

Shows the printable version of the PDS with a visible purpose watermark.

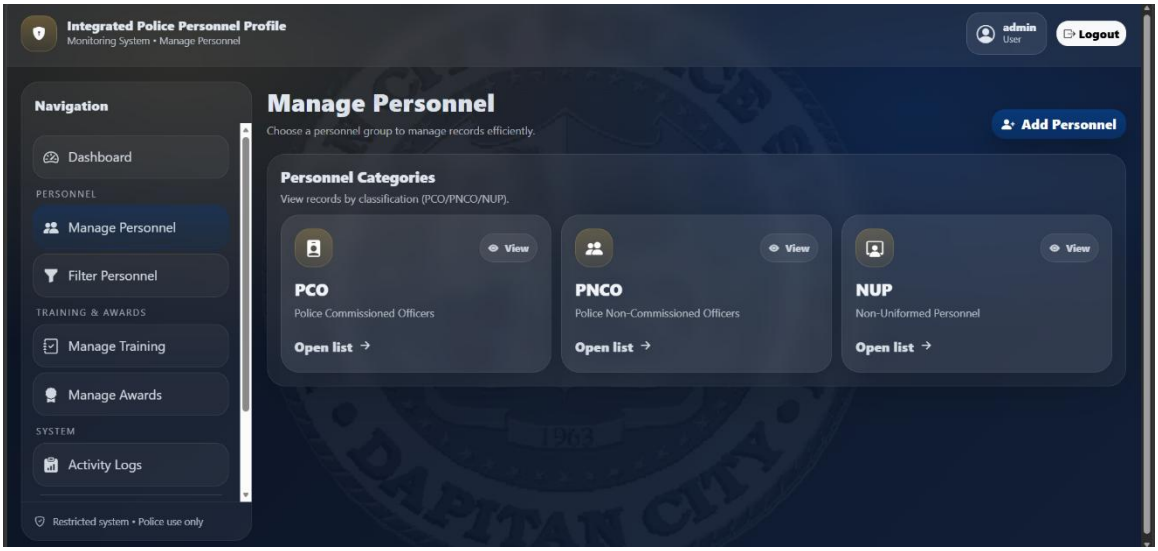


Figure 27. Manage Personnel Form

Used to view and organize personnel records by category.

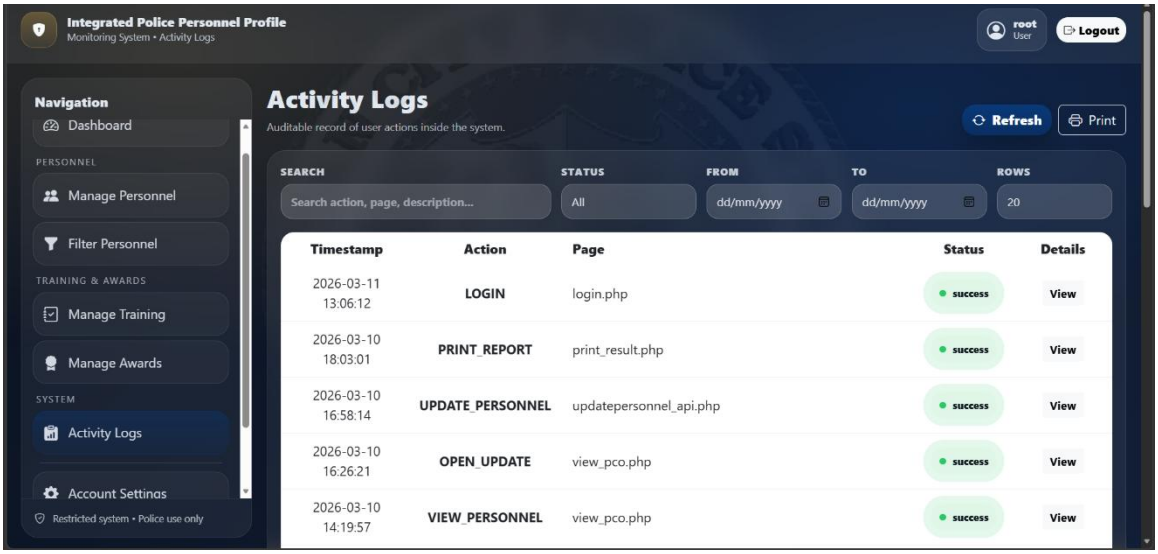


Figure 28. Activity Logs Form

shows a record of all actions performed by users in the system, including the date and time, type of action, accessed page, and status. It helps administrators track system activities and review detailed logs for monitoring and security purposes.



Figure 29. Account Setting Form

Allows the user to change their password by entering the current and new password.

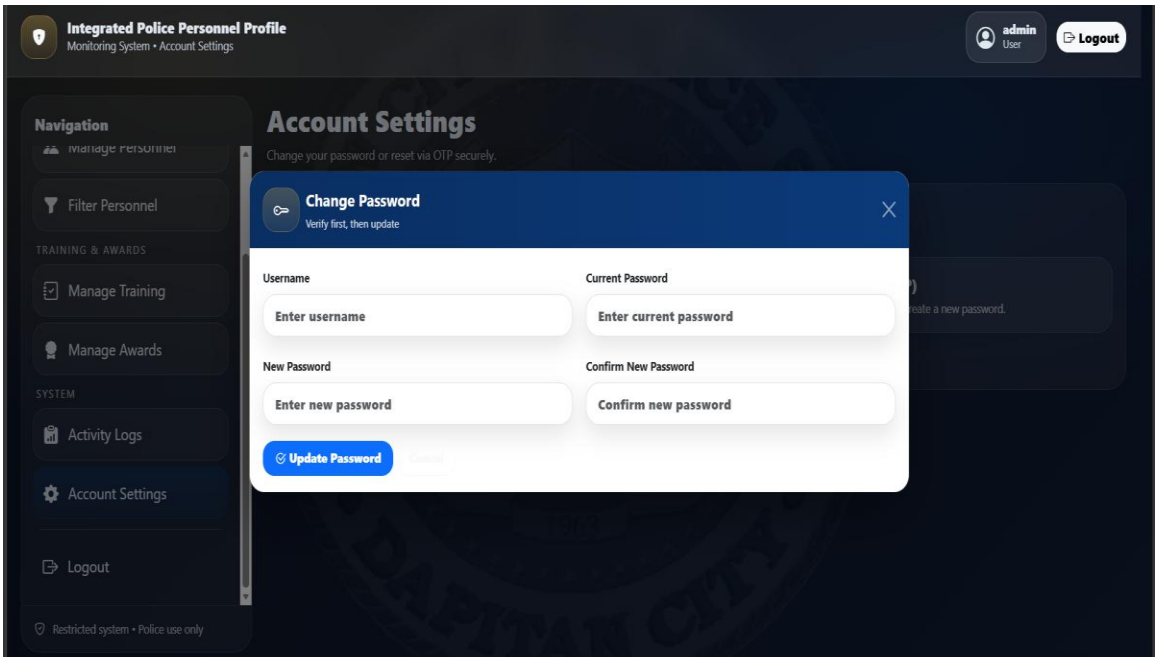


Figure 30. Change User Password Window

Used to manage and update the user account securely.

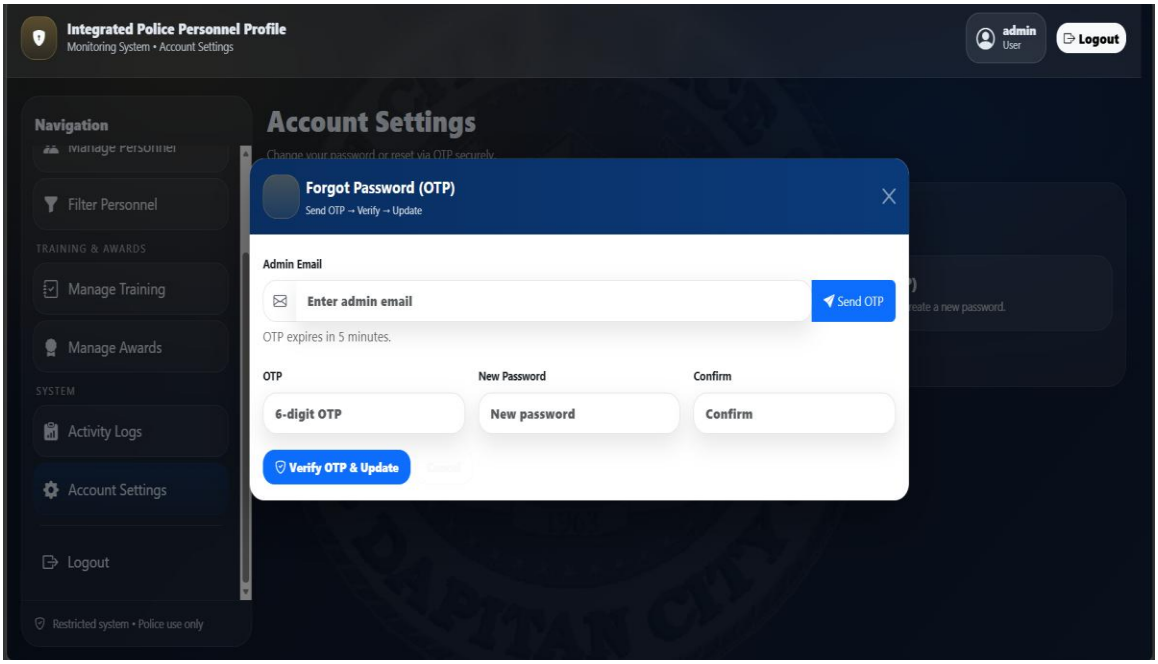


Figure 31. Forgot Password Window

Let the user reset their password through their registered email.

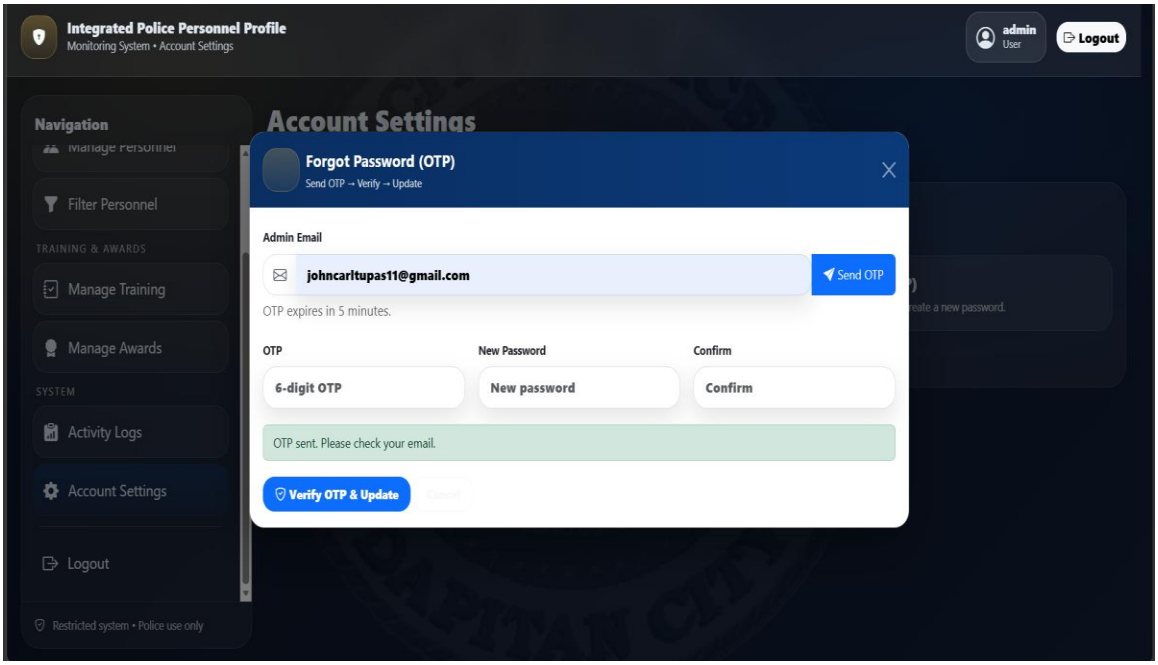


Figure 32. One-Time Password Verification Window

Lets the user reset their password using the one-time password sent to their registered email.

Reports

Integrated Personnel Profiling and Monitoring System (IPPM) automatically generates and organizes data summaries related to personnel, training, and awards. It allows the user to view, print, or export reports in a clear and standardized format, eliminating manual preparation and minimizing errors. The system retrieves real-time data from the centralized database, enabling filtered and accurate reports that support efficient decision-making and accountability.

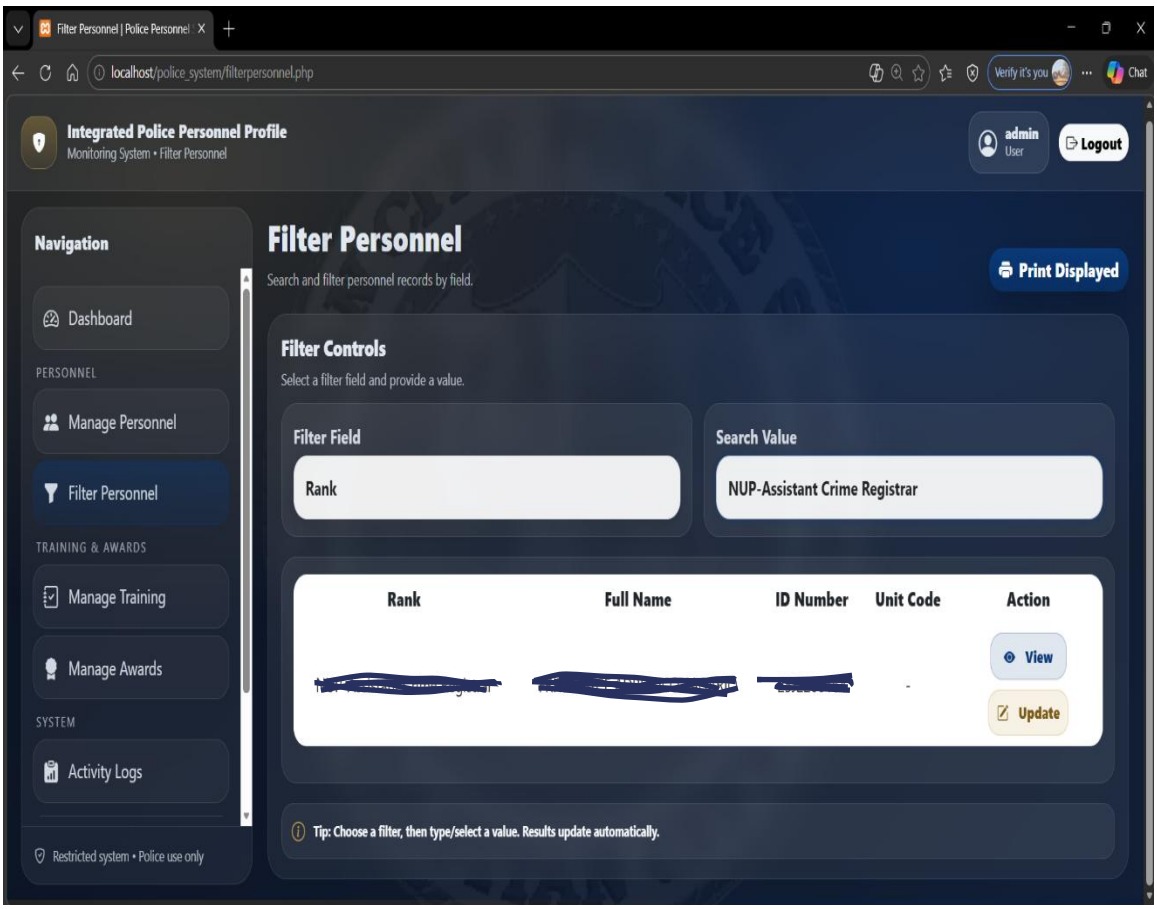


Figure 33. Generate Report – Rank

Displays a report generated by selecting a specific personnel rank, showing only the records that match the chosen filter.

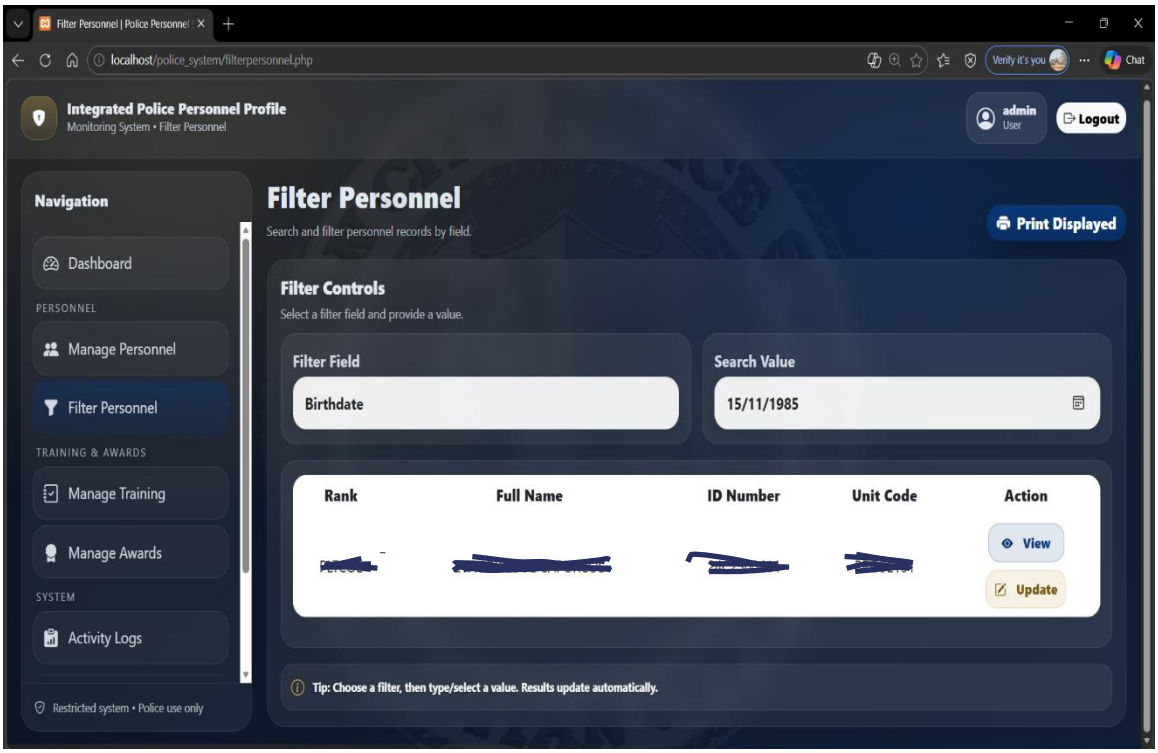


Figure 34. Generate Report - Birthdate

Displays a report generated by selecting a specific personnel birthdate.

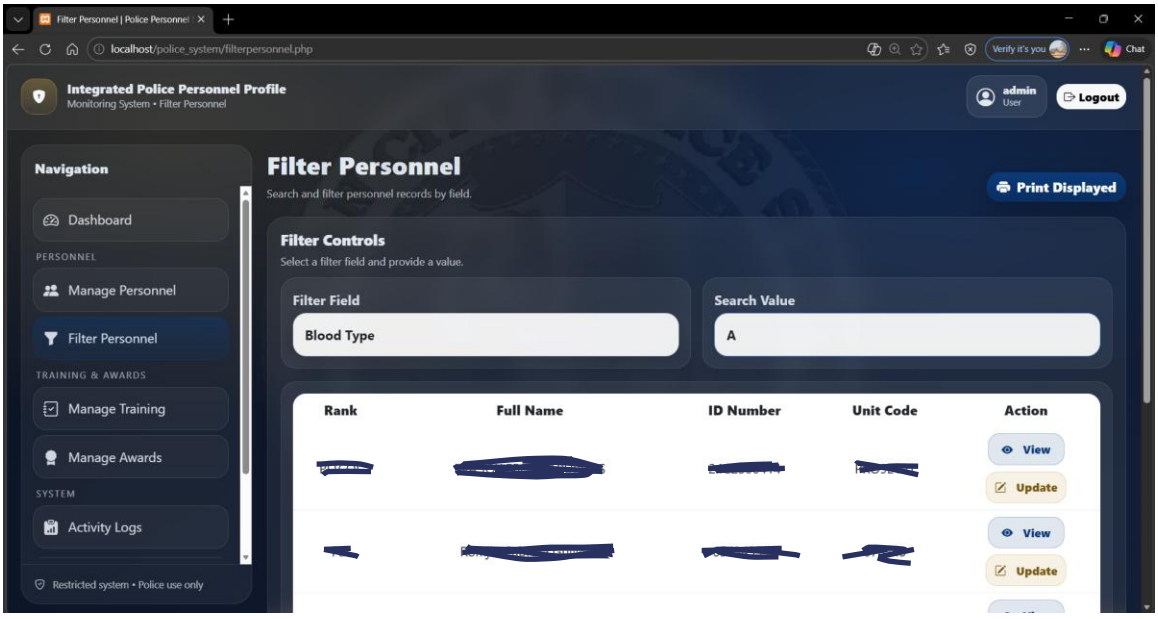


Figure 35. Generate Report – Blood Type

Displays a report generated by selecting a specific personnel blood type.

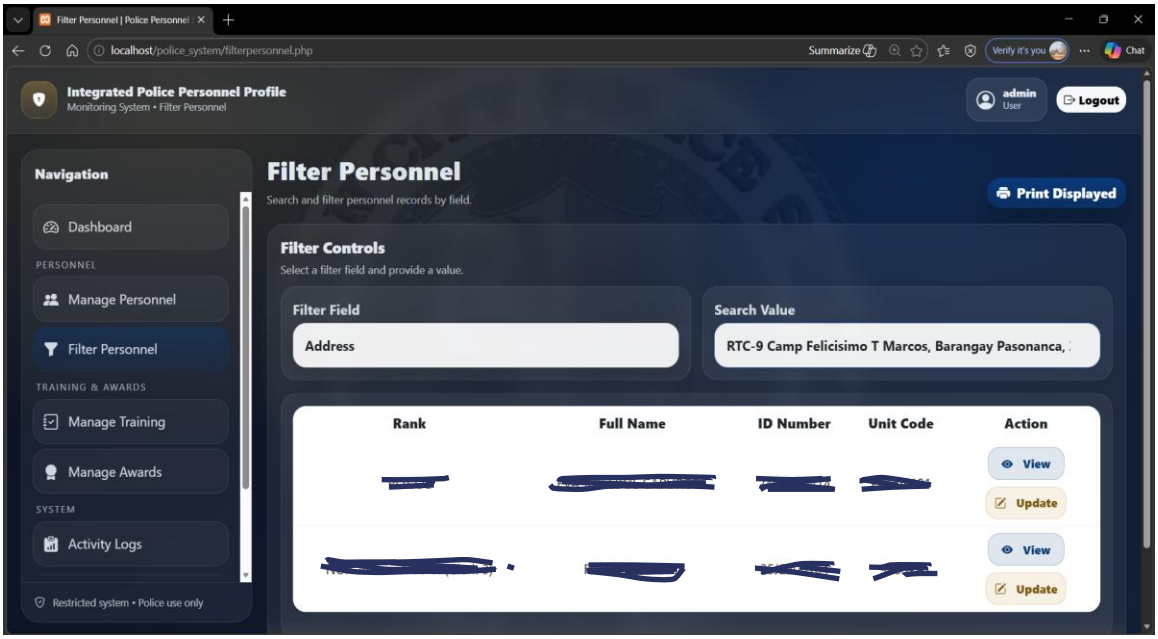


Figure 36. Generated Report through Address Filter

Displays a report generated by selecting a specific personnel address.

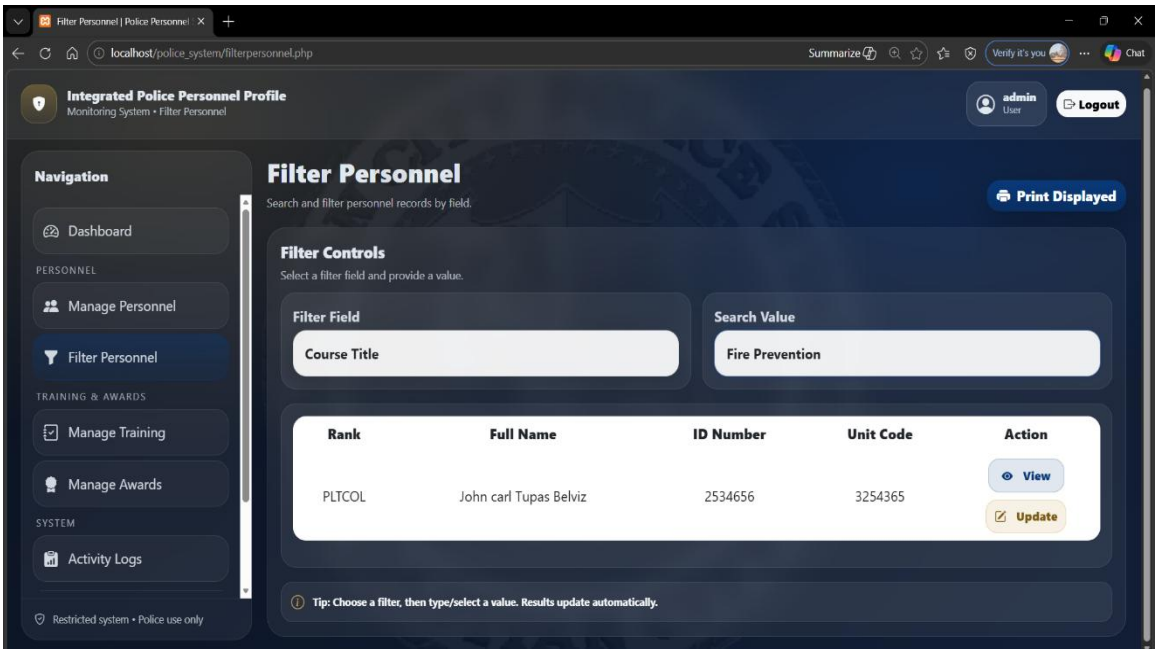


Figure 37. Generated Report through Course Title Filter

Displays a report generated by selecting a specific personnel attended training course title.

Gantt Chart

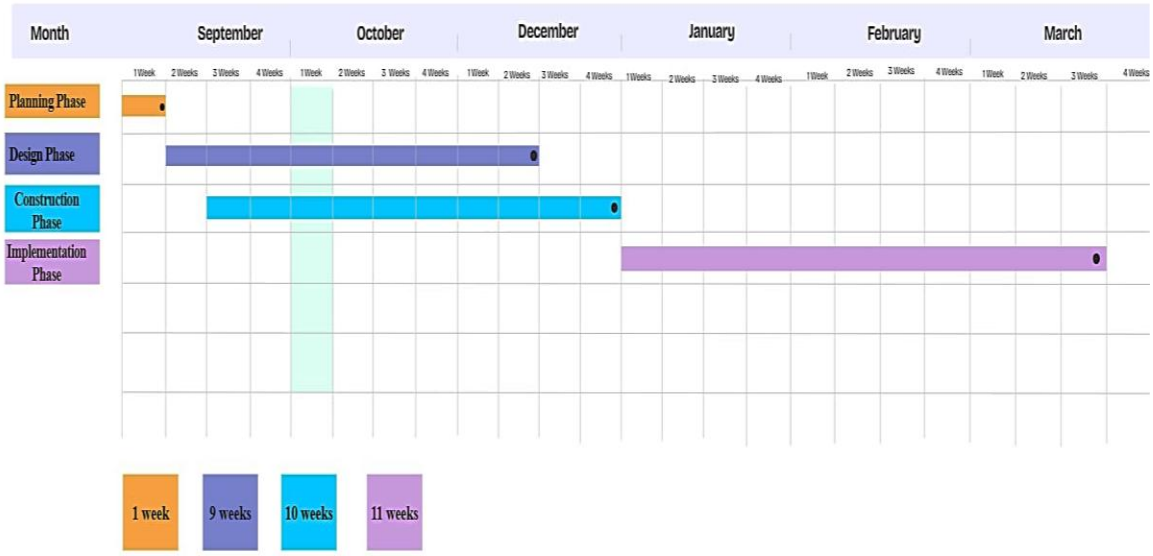


Figure 39. Gantt Chart on the Development of the IPPM

A Gantt Chart is a visual project management tool that shows the schedule and progress of each task in the project. It uses horizontal bars to represent the duration and timeline of activities.

Figure 39 shows the Gantt Chart for the development of the Integrated Personnel Profiling and Monitoring System (IPPM). It covers the Planning, Design, Construction, and Implementation Phases, scheduled from August, 2025 to March, 2026. This chart serves as a guide for tracking progress, managing deadlines, and ensuring timely completion of all project activities.